

Transverse pulses

7

Introduction and key concepts

 ESACF

This chapter forms the basis of the discussion into mechanical waves in the following chapters. We begin by discussing pulses. Pulses are disturbances in a medium. If you tap water in a bucket with your finger, notice that a ripple moves away from the point where you touched the water. The ripple is a pulse moving away from where you touched the water. See introductory video: (▶) Video: VPchw at www.everythingscience.co.za)

What is a medium?

 ESACG

A medium is the substance or material through which a pulse moves. The medium carries the pulse from one place to another. The medium does not create the pulse and the medium is not the pulse. Therefore the medium does not travel with the pulse as the pulse moves through it.

In each medium, the particles that make up the medium are moved *temporarily* from their rest position. In order for a pulse to travel, the different parts of the medium must be able to interact with each other.

A disturbance in water



Picture by [mikeyskatie](#) on Flickr.com

DEFINITION: Medium

A medium is the substance or material in which a pulse will move.

Pulses: amplitude and length

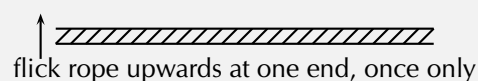


What is a pulse?



Investigation: Observation of pulses

Take a heavy rope. Have two people hold the rope stretched out horizontally. Flick the rope at one end only once.



What happens to the disturbance that you created in the rope? Does it stay at the place where it was created or does it move down the length of the rope?

In the activity, we created a *pulse*. A pulse is a *single* disturbance that moves through a medium. In a transverse pulse the displacement of the medium is perpendicular to the direction of motion of the pulse. Figure 7.2 shows an example of a transverse pulse. In the activity, the rope or spring was held horizontally and the pulse moved the rope up and down. This was an example of a transverse pulse.

DEFINITION: Pulse

A pulse is a single disturbance that moves through a medium.

DEFINITION: *Transverse Pulse*

A pulse where all of the particles disturbed by the pulse move perpendicular (at a right angle) to the direction in which the pulse is moving.

Pulse length and amplitude

The amplitude of a pulse is a measurement of how far the medium is displaced momentarily from a position of rest. The pulse length is a measurement of how long the pulse is. Both these quantities are shown in Figure 7.2.

DEFINITION: *Amplitude*

The amplitude of a pulse is the maximum disturbance or distance the medium is displaced from its rest (equilibrium) position.

Quantity: Amplitude (A)

Unit name: metre

Unit symbol: *m*

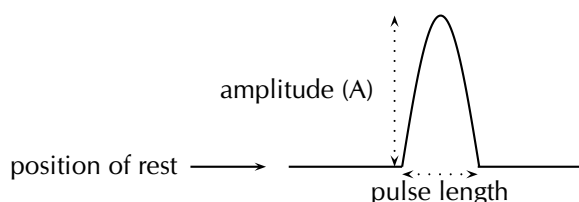
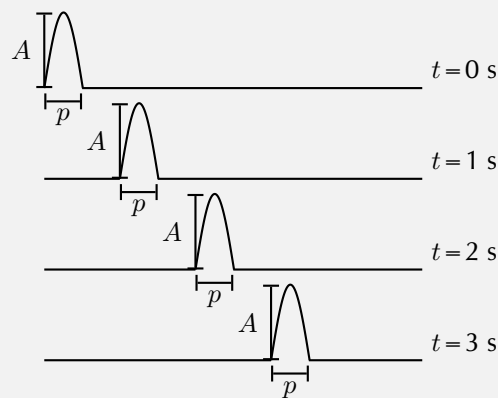


Figure 7.2: Example of a transverse pulse

The position of rest is the position the medium would be in if it were undisturbed. This is also called the equilibrium position. People will often use rest and equilibrium interchangeably.

Investigation: *Pulse length and amplitude*

The graphs below show the positions of a pulse at different times.



Use your ruler to measure the lengths of a and p . Fill your answers in the table.

Sign	Symbol	Meaning
Sign	Symbol	Meaning
Time	A	p
$t = 0\text{ s}$		
$t = 1\text{ s}$		
$t = 2\text{ s}$		
$t = 3\text{ s}$		

What do you notice about the values of A and p ?

In the activity, we found that the values for how high the pulse (A) is and how wide the pulse (p) is the same at different times. *Pulse length* and *amplitude* are two important quantities of a pulse.

Pulse speed**DEFINITION:** *Pulse speed*

Pulse speed is the distance a pulse travels per unit time.

Quantity: Pulse speed (v)

Unit name: metre per second

Unit

symbol: $\text{m} \cdot \text{s}^{-1}$

Speed is defined as the distance travelled per unit time (this will be covered in more detail in Motion in One Dimension). If the pulse travels a distance D in a time t , then the pulse speed v is:

$$v = \frac{D}{t}$$

Example 1: *Pulse speed***QUESTION**

A pulse covers a distance of 2 m in 4 s on a heavy rope. Calculate the pulse speed.

SOLUTION**Step 1 : Analyse the question**

We are given:

- the distance travelled by the pulse: $D = 2 \text{ m}$
- the time taken to travel 2 m: $t = 4 \text{ s}$

We are required to calculate the speed of the pulse.

Step 2 : Apply the relevant principles

We can use:

$$v = \frac{D}{t}$$

to calculate the speed of the pulse.

Step 3 : Do the calculation

$$\begin{aligned}
 v &= \frac{D}{t} \\
 &= \frac{2 \text{ m}}{4 \text{ s}} \\
 &= 0,5 \text{ m} \cdot \text{s}^{-1}
 \end{aligned}$$

Step 4 : **Quote the final result**

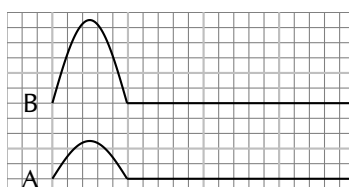
The pulse speed is $0,5 \text{ m} \cdot \text{s}^{-1}$.

Tip

The pulse speed depends on the properties of the medium and not on the amplitude or pulse length of the pulse.

Exercise 7 - 1

1. A pulse covers a distance of 5 m in 15 s. Calculate the speed of the pulse.
2. A pulse has a speed of $5 \text{ cm} \cdot \text{s}^{-1}$. How far does it travel in 2,5 s?
3. A pulse has a speed of $0,5 \text{ m} \cdot \text{s}^{-1}$. How long does it take to cover a distance of 25 cm?
4. How long will it take a pulse moving at $0,25 \text{ m} \cdot \text{s}^{-1}$ to travel a distance of 20 m?
5. The diagram shows two pulses in the same medium. Which has the higher speed? Explain your answer.



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(1.) 002q (2.) 002r (3.) 002s (4.) 002t (5.) 002u

Superposition of pulses



▶ See video: VPciu at www.everythingscience.co.za

Two or more pulses can pass through the same medium at that same time in the same place. When they do they interact with each other to form a different disturbance at that point. The resulting pulse is obtained by using the *principle of superposition*.

DEFINITION: Principle of superposition

The principle of superposition states that when two disturbance occupy the same space at the same time the resulting disturbance is the sum of two disturbances.

After pulses pass through each other, each pulse continues along its original direction of travel, and their original amplitudes remain unchanged.

Constructive interference takes place when two pulses meet each other to create a larger pulse. The amplitude of the resulting pulse is the sum of the amplitudes of the two initial pulses. This could be two crests meeting or two troughs meeting. This is shown in Figure 7.3.

DEFINITION: Constructive interference

Constructive interference is when two pulses meet, resulting in a bigger pulse.

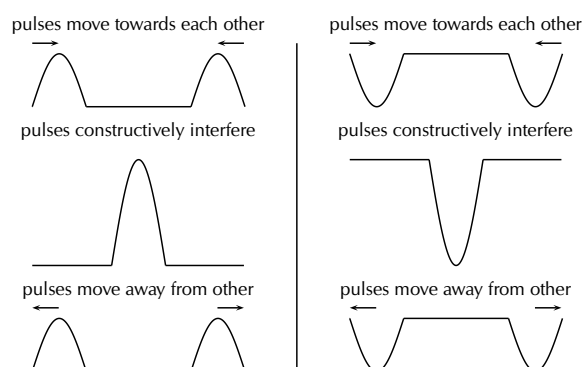


Figure 7.3: Superposition of two pulses: constructive interference.

Destructive interference takes place when two pulses meet and result in a smaller amplitude disturbance. The amplitude of the resulting pulse is the sum of the amplitudes of the two initial pulses, but the one amplitude will be a negative number. This is shown in Figure 7.4. In general, amplitudes of individual pulses are summed together to give the amplitude of the resultant pulse.

DEFINITION: *Destructive interference*

Destructive interference is when two pulses meet, resulting in a smaller pulse.

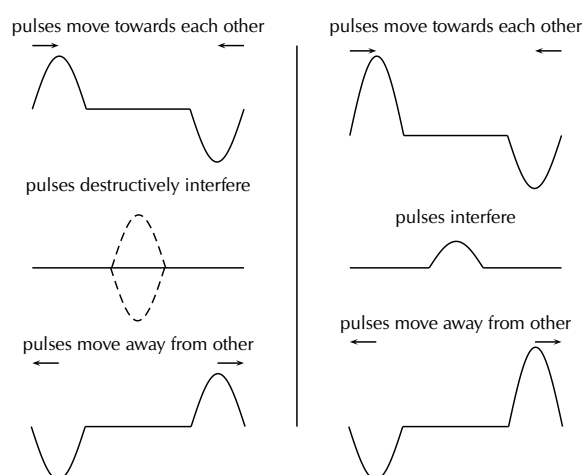
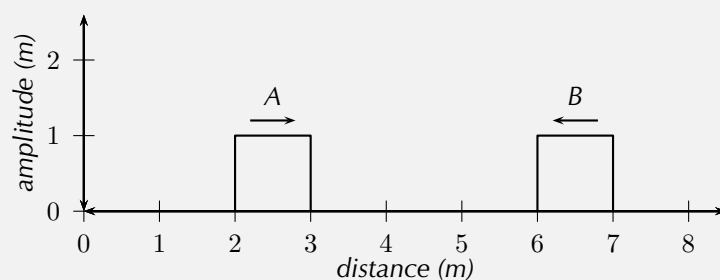


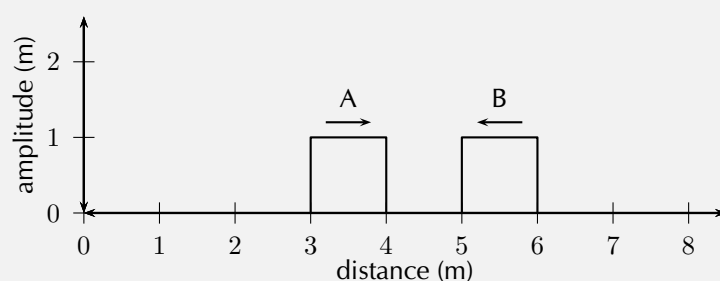
Figure 7.4: Superposition of two pulses. The left-hand series of images demonstrates destructive interference, since the pulses cancel each other. The right-hand series of images demonstrate a partial cancellation of two pulses, as their amplitudes are not the same in magnitude.

Example 2: Superposition of pulses**QUESTION**

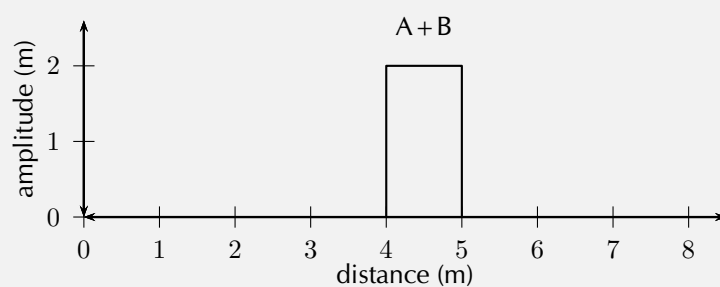
The two pulses shown below approach each other at $1 \text{ m} \cdot \text{s}^{-1}$. Draw what the waveform would look like after 1 s, 2 s and 5 s.

**SOLUTION****Step 1 : After 1 s**

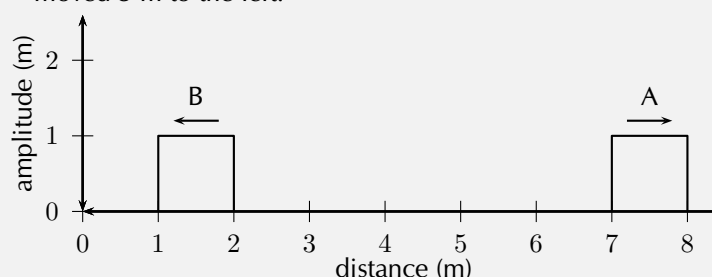
After 1 s, pulse A has moved 1 m to the right and pulse B has moved 1 m to the left.

**Step 2 : After 2 s**

After 1 s more, pulse A has moved 1 m to the right and pulse B has moved 1 m to the left.

**Step 3 : After 5 s**

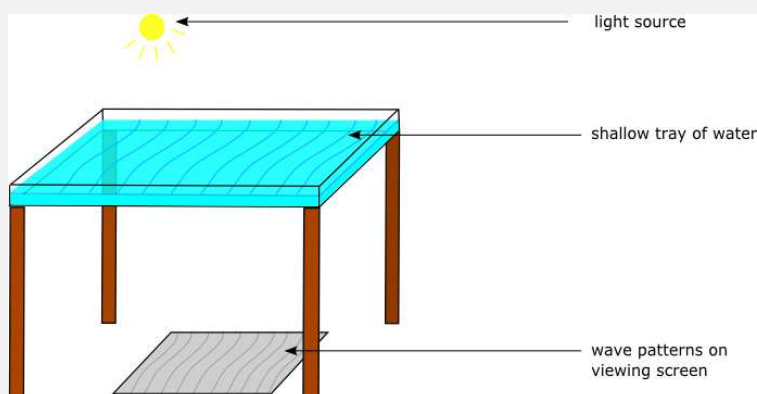
After 5 s more, pulse A has moved 5 m to the right and pulse B has moved 5 m to the left.



General experiment: *Constructive and destructive interference*

Aim: To demonstrate constructive and destructive interference

Apparatus: Ripple tank apparatus



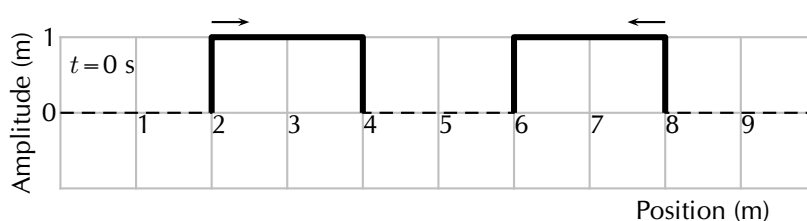
Method:

1. Set up the ripple tank
2. Produce a single pulse and observe what happens (you can do this any means, tapping the water with a finger, dropping a small object into the water, tapping a ruler or even using a electronic vibrator)
3. Produce two pulses simultaneously and observe what happens
4. Produce two pulses at slightly different times and observe what happens

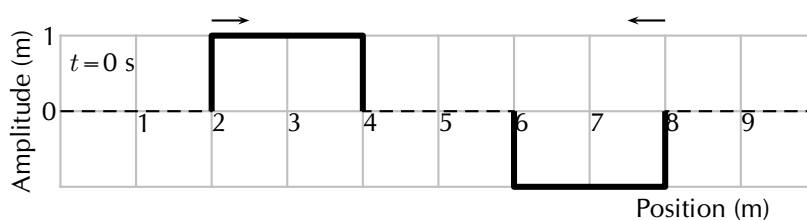
Results and conclusion: You should observe that when you produce two pulses simultaneously you see them interfere constructively and when you produce two pulses at slightly different times you see them interfere destructively.

Exercise 7 - 2

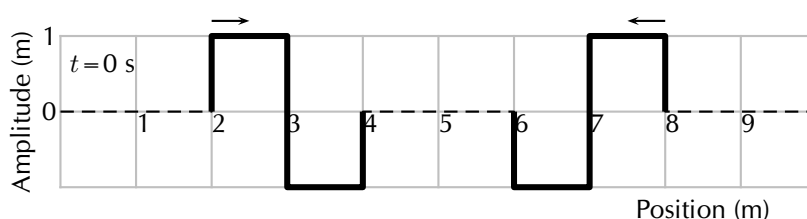
1. For the following pulse, draw the resulting wave forms after 1 s, 2 s, 3 s, 4 s and 5 s. Each pulse is travelling at $1 \text{ m} \cdot \text{s}^{-1}$. Each block represents 1 m. The pulses are shown as thick black lines and the undisplaced medium as dashed lines.



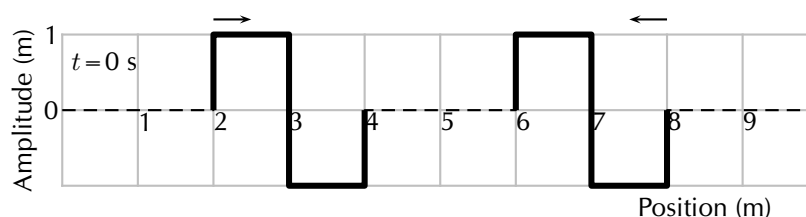
2. For the following pulse, draw the resulting wave forms after 1 s, 2 s, 3 s, 4 s and 5 s. Each pulse is travelling at $1 \text{ m} \cdot \text{s}^{-1}$. Each block represents 1 m. The pulses are shown as thick black lines and the undisplaced medium as dashed lines.



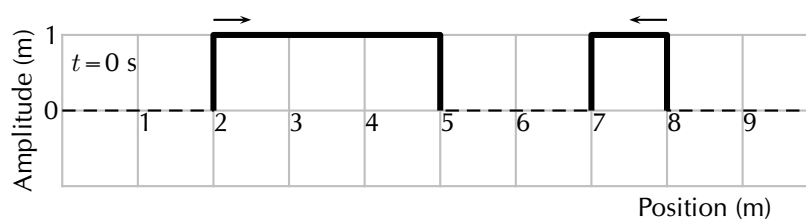
3. For the following pulse, draw the resulting wave forms after 1 s, 2 s, 3 s, 4 s and 5 s. Each pulse is travelling at $1 \text{ m} \cdot \text{s}^{-1}$. Each block represents 1 m. The pulses are shown as thick black lines and the undisplaced medium as dashed lines.



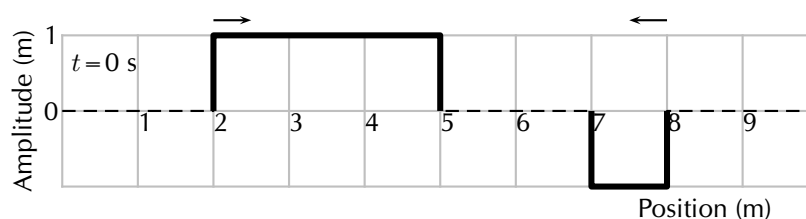
4. For the following pulse, draw the resulting wave forms after 1 s, 2 s, 3 s, 4 s and 5 s. Each pulse is travelling at $1 \text{ m} \cdot \text{s}^{-1}$. Each block represents 1 m. The pulses are shown as thick black lines and the undisplaced medium as dashed lines.



5. For the following pulse, draw the resulting wave forms after 1 s, 2 s, 3 s, 4 s and 5 s. Each pulse is travelling at $1 \text{ m} \cdot \text{s}^{-1}$. Each block represents 1 m. The pulses are shown as thick black lines and the undisplaced medium as dashed lines.



6. For the following pulse, draw the resulting wave forms after 1 s, 2 s, 3 s, 4 s and 5 s. Each pulse is travelling at $1 \text{ m} \cdot \text{s}^{-1}$. Each block represents 1 m. The pulses are shown as thick black lines and the undisplaced medium as dashed lines.



7. What is superposition of waves?
8. What is constructive interference?
9. What is destructive interference?

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- (1.) 002v (2.) 002w (3.) 002x (4.) 002y (5.) 002z (6.) 0030
 (7.) 0031 (8.) 0032 (9.) 0033

Chapter 7 | Summary

See the summary presentation (📺 Presentation: VPcjo at www.everythingscience.co.za)

- A medium is the substance or material in which a pulse will move.
- A pulse is a single disturbance that moves through a medium.
- The amplitude of a pulse is the maximum disturbance or distance the medium is displaced from its equilibrium position (rest).
- Pulse speed is the distance a pulse travels per unit time.
- Constructive interference is when two pulses meet and result in a bigger pulse.
- Destructive interference is when two pulses meet and result in a smaller pulse.

Physical Quantities		
Quantity	Unit name	Unit symbol
Amplitude (A)	metre	m
Pulse speed (v)	metre per second	$\text{m} \cdot \text{s}^{-1}$

Table 7.1: Units used in **transverse pulses**

Chapter 7

End of chapter exercises

1. A heavy rope is flicked upwards, creating a single pulse in the rope. Make a drawing of the rope and indicate the following in your drawing:
 - a. The direction of motion of the pulse

- b. Amplitude
 - c. Pulse length
 - d. Position of rest
2. A pulse has a speed of $2,5 \text{ m} \cdot \text{s}^{-1}$. How far will it have travelled in 6 s?
3. A pulse covers a distance of 75 cm in 2,5 s. What is the speed of the pulse?
4. How long does it take a pulse to cover a distance of 200 mm if its speed is $4 \text{ m} \cdot \text{s}^{-1}$?

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(1.) 0034 (2.) 0035 (3.) 0036 (4.) 0037

Transverse waves

8

Introduction

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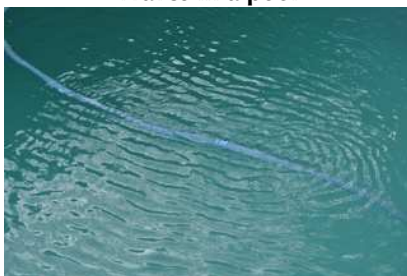
Waves occur frequently in nature. The most obvious examples are waves in water, on a dam, in the ocean, or in a bucket. We are interested in the properties that waves have. All waves have the same properties.

Waves do not only occur in water, they occur in any kind of medium. Earthquakes release enough energy to create waves that are powerful enough to travel through the rock of the Earth. When your friend speaks to you *sound* waves are produced that travel through the air to your ears. A wave is simply the disturbance of a medium by moving energy but how is it different from a pulse?. See introductory video: (▶ Video: VPgio at www.everythingscience.co.za)

What is a transverse wave?

www ESACL

Waves in a pool



We have studied pulses in *Transverse pulses*, and know that a pulse is a single disturbance that travels through a medium. A wave is a periodic, continuous disturbance that consists of a *train* or *succession* of pulses.

An enlarged version of the ripple tank can be seen in a real life example of a Kreepy Krauly® making waves in a pool because of the regular vibrations. The Kreepy Krauly® was invented in South Africa by Ferdinand Chauvier and his son Daniel.

DEFINITION: Wave

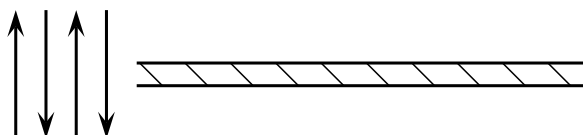
A wave is a periodic, continuous disturbance that consists of a *train* of pulses.

DEFINITION: *Transverse wave*

A *transverse wave* is a wave where the movement of the particles of the medium is perpendicular (at a right angle) to the direction of propagation of the wave.

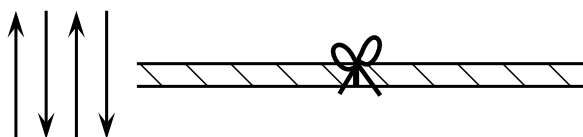
Activity:*Transverse waves*

Take a rope or slinky spring. Have two people hold the rope or spring stretched out horizontally. Flick the one end of the rope up and down **continuously** to create a *train of pulses*.



Flick rope up and down

1. Describe what happens to the rope.
2. Draw a diagram of what the rope looks like while the pulses travel along it.
3. In which direction do the pulses travel?
4. Tie a ribbon to the middle of the rope. This indicates a particle in the rope.



Flick rope up and down

5. Flick the rope continuously. Watch the ribbon carefully as the pulses travel through the rope. What happens to the ribbon?
6. Draw a picture to show the motion of the ribbon. Draw the ribbon as a dot and use arrows to indicate how it moves.

In the activity, you created waves. The medium through which these waves propagated was the rope, which is obviously made up of a very large number of particles (atoms). From the activity, you would have noticed that the wave travelled from one side to the other, but the particles (the ribbon) moved only up and down.

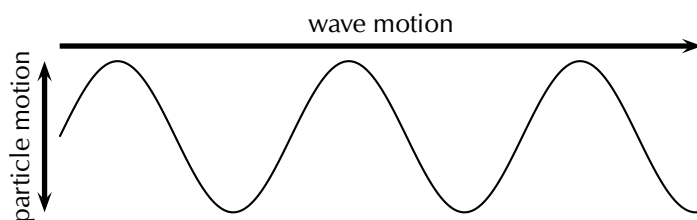


Figure 8.2: A transverse wave, showing the direction of motion of the wave perpendicular to the direction in which the particles move.

When the particles of a medium move at right angles to the direction of propagation of a wave, the wave is called *transverse*. For waves, there is no net displacement of the particles of the medium (they return to their equilibrium position), but there is a net displacement of the wave. There are thus two different motions: the motion of the particles of the medium and the motion of the wave.

The following simulation will help you understand more about waves. Select the oscillate option and then observe what happens. See simulation: (📺) Simulation: VPnue at www.everythingscience.co.za

Crests and troughs



Waves have moving *crests* (or *peaks*) and *troughs*. A crest is the highest point the medium rises to and a trough is the lowest point the medium sinks to.

crests and troughs on a transverse wave are shown in Figure 8.3.

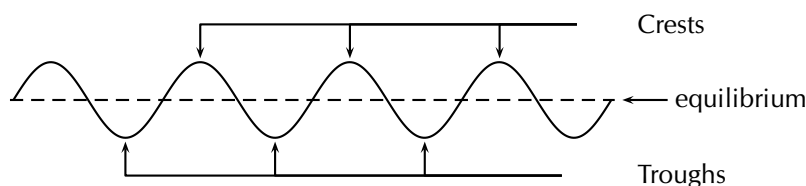


Figure 8.3: Crests and troughs in a transverse wave.

DEFINITION: Crests and troughs

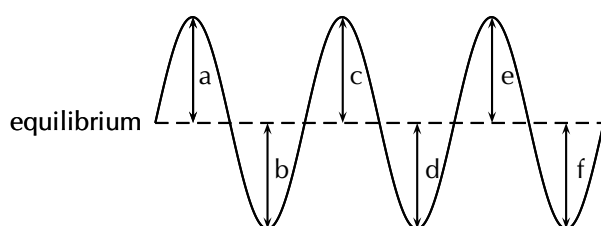
A *crest* is a point on the wave where the displacement of the medium is at a maximum. A point on the wave is a *trough* if the displacement of the medium at that point is at a minimum.

Amplitude



Activity:

Amplitude



Fill in the table below by measuring the distance between the equilibrium and each crest and trough in the wave above. Use your ruler to measure the distances.

Crest/Trough	Measurement (cm)
a	
b	
c	
d	
e	
f	

1. What can you say about your results?
2. Are the distances between the equilibrium position and each crest equal?
3. Are the distances between the equilibrium position and each trough equal?
4. Is the distance between the equilibrium position and crest equal to the distance between equilibrium and trough?

As we have seen in the activity on amplitude, the distance between the crest and the equilibrium position is equal to the distance between the trough and the equilibrium position. This distance is known as the *amplitude* of the wave, and is the characteristic height of the wave, above or below the equilibrium position. Normally the symbol A is used to represent the amplitude of a wave. The SI unit of amplitude is the metre (m).

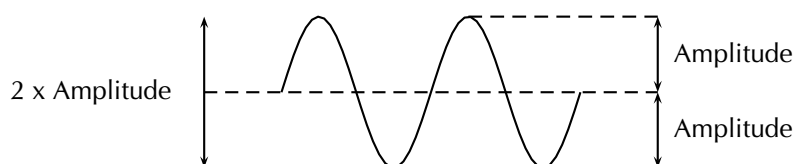
DEFINITION: Amplitude

The amplitude of a wave is the maximum disturbance or displacement of the medium from the equilibrium (rest) position.

Quantity: Amplitude (A)

Unit name: metre

Unit symbol: m

**Example 1: Amplitude of sea waves****QUESTION**

If the crest of a wave measures 2 m above the still water mark in the harbour, what is the amplitude of the wave?

SOLUTION**Step 1 : Analyse the information provided**

We have been told that the harbour has a still water mark. This is a line created when there are no disturbances in the water, which means that it is the equilibrium position of the water.

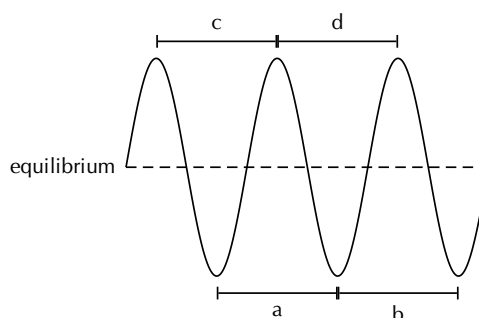
Step 2 : Determine the amplitude

The definition of the amplitude is the height of a crest above the equilibrium position. The still water mark is the height of the water at equilibrium and the crest is 2 m above this, so the amplitude is 2 m.

FACT

A tsunami is a series of sea waves caused by an underwater earthquake, landslide, or volcanic eruption. When the ocean is deep, tsunamis may be less than 30 cm high on the ocean's surface and can travel at speeds up to $700 \text{ km} \cdot \text{hr}^{-1}$. In shallow water near the coast, it slows down. The top of the wave moves faster than the bottom, causing the sea to rise dramatically, as much as 30 m. The wavelength can be as long as 100 km and the period as long as a hour.

In 2004, the Indian Ocean tsunami was caused by an earthquake that is thought to have had the energy of 23,000 atomic bombs. Within hours of the earthquake, killer waves radiating from away from the earthquake crashed into the coastline of 11 countries, killing 150,000 people. The final death toll was 283,000.

Activity:**Wavelength**

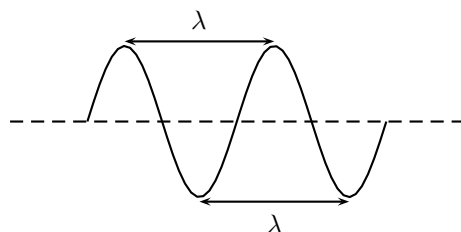
Fill in the table below by measuring the distance between crests and troughs in the wave above.

	Distance(cm)
a	
b	
c	
d	

1. What can you say about your results?
2. Are the distances between crests equal?
3. Are the distances between troughs equal?
4. Is the distance between crests equal to the distance between troughs?

As we have seen in the activity on wavelength, the distance between two *adjacent* crests is the same no matter which two adjacent crests you choose. There is a fixed distance between the crests. Similarly, we have seen that there is a fixed distance between the troughs, no matter which two troughs you look at. More importantly, the distance between two adjacent crests is the same as the distance between two adjacent troughs. This distance is called the *wavelength* of the wave.

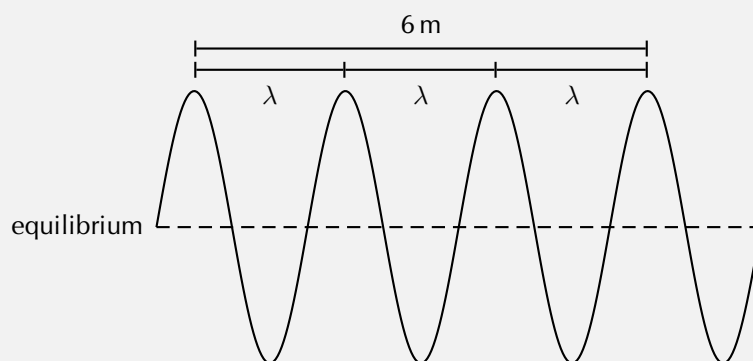
The symbol for the wavelength is λ (the Greek letter *lambda*) and wavelength is measured in metres (m).

**Example 2: Wavelength****QUESTION**

The total distance between 4 consecutive crests of a transverse wave is 6 m. What is the wavelength of the wave?

SOLUTION

Step 1 : **Draw a rough sketch of the situation**



Step 2 : **Determine how to approach the problem**

From the sketch we see that 4 consecutive crests is equivalent to 3 wavelengths.

Step 3 : **Solve the problem**

Therefore, the wavelength of the wave is:

$$\begin{aligned} 3\lambda &= 6 \text{ m} \\ \lambda &= \frac{6 \text{ m}}{3} \\ &= 2 \text{ m} \end{aligned}$$

Step 4 : **Quote the final answer**

The wavelength is 2 m.

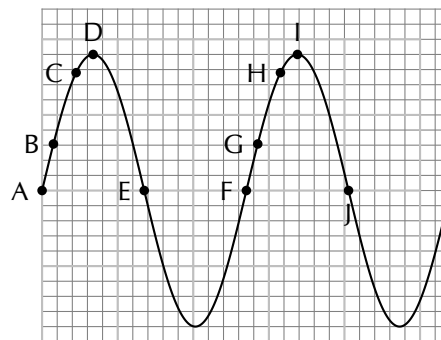
Points in phase



Activity:

Points in phase

Fill in the table by measuring the distance between the indicated points.



Points	Distance (cm)
A to F	
B to G	
C to H	
D to I	
E to J	

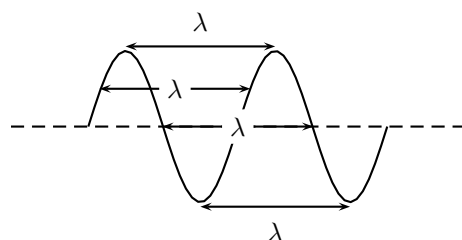
What do you find?

In the activity the distance between the indicated points was equal. These points are then said to be *in phase*. Two points in phase are separated by a whole (1, 2, 3, ...) number multiple of whole wave cycles or wavelengths. The points in phase do not have to be crests or troughs, but they must be separated by a complete number of wavelengths.

We then have an alternate definition of the wavelength as the distance between any two adjacent points which are *in phase*.

DEFINITION: *Wavelength of wave*

The wavelength of a wave is the distance between any two adjacent points that are in phase.



Points that are not in phase, those that are not separated by a complete number of wavelengths, are called *out of phase*. Examples of points like these would be *A* and *C*, or *D* and *E*, or *B* and *H* in the Activity.

Period and frequency

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Imagine you are sitting next to a pond and you watch the waves going past you. First one crest arrives, then a trough, and then another crest. Suppose you measure the time taken between one crest arriving and then the next. This time will be the same for any two successive crests passing you. We call this time the *period*, and it is a characteristic of the wave.

The symbol T is used to represent the period. The period is measured in seconds (s).

DEFINITION: *Period*

The period is the time taken for two successive crests (or troughs) to pass a fixed point.

Quantity: Period (T)

Unit name: second

Unit symbol: s

Imagine the pond again. Just as a crest passes you, you start your stopwatch and count each crest going past. After 1 second you stop the clock and stop counting. The number of

crests that you have counted in the 1 second is the *frequency* of the wave.

DEFINITION: Frequency

The frequency is the number of successive crests (or troughs) passing a given point in 1 second.

Quantity: Frequency (f)

Unit name: hertz

Unit symbol: Hz

The frequency and the period are related to each other. As the period is the time taken for 1 crest to pass, then the number of crests passing the point in 1 second is $\frac{1}{T}$. But this is the frequency. So

$$f = \frac{1}{T}$$

or alternatively,

$$T = \frac{1}{f}$$

For example, if the time between two consecutive crests passing a fixed point is $\frac{1}{2}$ s, then the period of the wave is $\frac{1}{2}$ s. Therefore, the frequency of the wave is:

$$\begin{aligned} f &= \frac{1}{T} \\ &= \frac{1}{\frac{1}{2} \text{ s}} \\ &= 2 \text{ s}^{-1} \end{aligned}$$

The unit of frequency is the Hertz (Hz) or s^{-1} .

Example 3: Period and frequency

QUESTION

What is the period of a wave of frequency 10 Hz?

SOLUTION

Step 1 : **Determine what is given and what is required**

We are required to calculate the period of a 10 Hz wave.

Step 2 : **Determine how to approach the problem**

We know that:

$$T = \frac{1}{f}$$

Step 3 : **Solve the problem**

$$\begin{aligned} T &= \frac{1}{f} \\ &= \frac{1}{10 \text{ Hz}} \\ &= 0,1 \text{ s} \end{aligned}$$

Step 4 : **Write the answer**

The period of a 10 Hz wave is 0,1 s.

Speed of a transverse wave



DEFINITION: Wave speed

Wave speed is the distance a wave travels per unit time.

Quantity: Wave speed (v)

Unit name: metre per second

Unit

symbol: $\text{m} \cdot \text{s}^{-1}$

The distance between two successive crests is 1 wavelength, λ . Thus in a time of 1 period, the wave will travel 1 wavelength in distance. Thus the speed of the wave, v , is:

$$v = \frac{\text{distance travelled}}{\text{time taken}} = \frac{\lambda}{T}$$

However, $f = \frac{1}{T}$. Therefore, we can also write:

$$\begin{aligned} v &= \frac{\lambda}{T} \\ &= \lambda \cdot \frac{1}{T} \\ &= \lambda \cdot f \end{aligned}$$

We call this equation the *wave equation*. To summarise, we have that $v = \lambda \cdot f$ where

- v = speed in $\text{m} \cdot \text{s}^{-1}$
- λ = wavelength in m
- f = frequency in Hz

Wave equation

$$v = f \cdot \lambda$$

or

$$v = \frac{\lambda}{T}$$

Example 4: Speed of a transverse wave I

QUESTION

When a particular string is vibrated at a frequency of 10 Hz, a transverse wave of wavelength 0,25 m is produced. Determine the speed of the wave as it travels along the string.

SOLUTION

Step 1 : **Determine what is given and what is required**

- frequency of wave: $f = 10 \text{ Hz}$
- wavelength of wave: $\lambda = 0,25 \text{ m}$

We are required to calculate the speed of the wave as it travels along the string.

All quantities are in SI units.

Step 2 : **Determine how to approach the problem**

We know that the speed of a wave is:

$$v = f \cdot \lambda$$

and we are given all the necessary quantities.

Step 3 : **Substituting in the values**

$$\begin{aligned}
 v &= f \cdot \lambda \\
 &= (10 \text{ Hz})(0,25 \text{ m}) \\
 &= (10 \text{ s}^{-1})(0,25 \text{ m}) \\
 &= 2,5 \text{ m} \cdot \text{s}^{-1}
 \end{aligned}$$

Step 4 : **Write the final answer**

The wave travels at $2,5 \text{ m} \cdot \text{s}^{-1}$ along the string.

Example 5: Speed of a transverse wave II

QUESTION

A cork on the surface of a swimming pool bobs up and down once every second on some ripples. The ripples have a wavelength of 20 cm. If the cork is 2 m from the edge of the pool, how long does it take a ripple passing the cork to reach the edge?

SOLUTION

Step 1 : **Determine what is given and what is required**

We are given:

- frequency of wave: $f = 1 \text{ Hz}$
- wavelength of wave: $\lambda = 20 \text{ cm}$
- distance of cork from edge of pool: $D = 2 \text{ m}$

We are required to determine the time it takes for a ripple to travel between the cork and the edge of the pool.

The wavelength is not in SI units and should be converted.

Step 2 : **Determine how to approach the problem**

The time taken for the ripple to reach the edge of the pool is obtained from:

$$t = \frac{D}{v} \quad \left(\text{from } v = \frac{D}{t} \right)$$

We know that

$$v = f \cdot \lambda$$

Therefore,

$$t = \frac{D}{f \cdot \lambda} \quad (8.1)$$

Step 3 : **Convert wavelength to SI units**

$$20 \text{ cm} = 0,2 \text{ m}$$

Step 4 : **Solve the problem**

$$\begin{aligned} t &= \frac{d}{f \cdot \lambda} \\ &= \frac{2 \text{ m}}{(1 \text{ Hz})(0,2 \text{ m})} \\ &= \frac{2 \text{ m}}{(1 \text{ s}^{-1})(0,2 \text{ m})} \\ &= 10 \text{ s} \end{aligned}$$

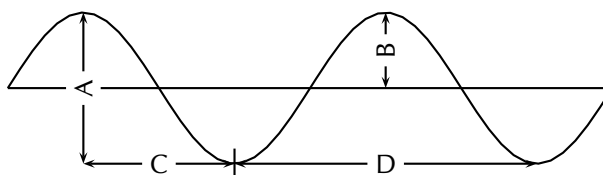
Step 5 : **Write the final answer**

A ripple passing the leaf will take 10 s to reach the edge of the pool.

► See video: VPcci at www.everythingscience.co.za

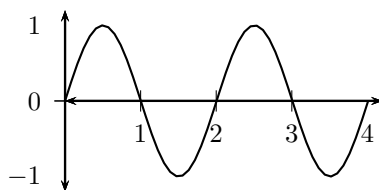
Exercise 8 - 1

1. When the particles of a medium move perpendicular to the direction of the wave motion, the wave is called a wave.
2. A transverse wave is moving downwards. In what direction do the particles in the medium move?
3. Consider the diagram below and answer the questions that follow:



- a. the wavelength of the wave is shown by letter _____.

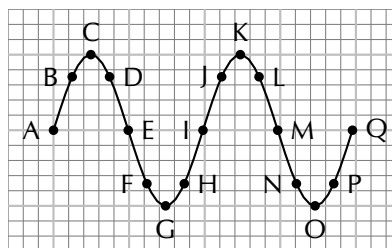
- b. the amplitude of the wave is shown by letter _____.
4. Draw 2 wavelengths of the following transverse waves on the same graph paper. Label all important values.
- Wave 1: Amplitude = 1 cm, wavelength = 3 cm
 - Wave 2: Peak to trough distance (vertical) = 3 cm, crest to crest distance (horizontal) = 5 cm
5. You are given the transverse wave below.



Draw the following:

- A wave with twice the amplitude of the given wave.
 - A wave with half the amplitude of the given wave.
 - A wave travelling at the same speed with twice the frequency of the given wave.
 - A wave travelling at the same speed with half the frequency of the given wave.
 - A wave with twice the wavelength of the given wave.
 - A wave with half the wavelength of the given wave.
 - A wave travelling at the same speed with twice the period of the given wave.
 - A wave travelling at the same speed with half the period of the given wave.
6. A transverse wave travelling at the same speed with an amplitude of 5 cm has a frequency of 15 Hz. The horizontal distance from a crest to the nearest trough is measured to be 2,5 cm. Find the
- period of the wave.
 - speed of the wave.
7. A fly flaps its wings back and forth 200 times each second. Calculate the period of a wing flap.
8. As the period of a wave increases, the frequency **increases/decreases/does not change**.
9. Calculate the frequency of rotation of the second hand on a clock.
10. Microwave ovens produce radiation with a frequency of 2 450 MHz (1 MHz = 10^6 Hz) and a wavelength of 0,122 m. What is the wave speed of the radiation?

11. Study the following diagram and answer the questions:



- Identify two sets of points that are in phase.
 - Identify two sets of points that are out of phase.
 - Identify any two points that would indicate a wavelength.
12. Tom is fishing from a pier and notices that four wave crests pass by in 8 s and estimates the distance between two successive crests is 4 m. The timing starts with the first crest and ends with the fourth. Calculate the speed of the wave.

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(1.) 0039 (2.) 003a (3.) 003b (4.) 003c (5.) 003d (6.) 003e
 (7.) 003f (8.) 003g (9.) 003h (10.) 003i (11.) 003j (12.) 003k

Chapter 8 | Summary

See the summary presentation (Presentation: VPguj at www.everythingscience.co.za)

- A wave is formed when a continuous number of pulses are transmitted through a medium.
- A crest is the highest point a particle in the medium rises to.
- A trough is the lowest point a particle in the medium sinks to.
- In a transverse wave, the particles move perpendicular to the motion of the wave.
- The amplitude (A) is the maximum distance from equilibrium position to a crest (or trough), or the maximum displacement of a particle in a wave from its position of rest.

- The wavelength (λ) is the distance between any two adjacent points on a wave that are in phase. It is measured in metres.
- The period (T) of a wave is the time it takes a wavelength to pass a fixed point. It is measured in seconds (s).
- The frequency (f) of a wave is how many waves pass a point in a second. It is measured in hertz (Hz) or s^{-1} .
- Frequency: $f = \frac{1}{T}$
- Period: $T = \frac{1}{f}$
- Speed: $v = f \cdot \lambda$ or $v = \frac{\lambda}{T}$.

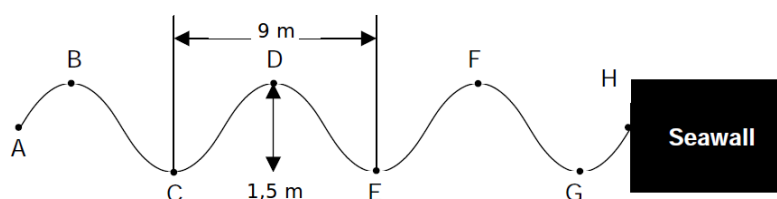
Physical Quantities		
Quantity	Unit name	Unit symbol
Amplitude (A)	metre	m
Wavelength (λ)	metre	m
Period (T)	second	s
Frequency (f)	hertz	Hz (s^{-1})
Wave speed (v)	metre per second	$m \cdot s^{-1}$

Table 8.1: Units used in **transverse waves**

Chapter 8

End of chapter exercises

1. A wave travels along a string at a speed of $1,5 m \cdot s^{-1}$. If the frequency of the source of the wave is 7,5 Hz, calculate:
 - a. the wavelength of the wave
 - b. the period of the wave
2. Water waves crash against a seawall around the harbour. Eight waves hit the seawall in 5 s. The distance between successive troughs is 9 m. The height of the waveform trough to crest is 1,5 m.



- a. How many complete waves are indicated in the sketch?

- b. Write down the letters that indicate any TWO points that are:
 - i. in phase
 - ii. out of phase
 - iii. Represent ONE wavelength.
- c. Calculate the amplitude of the wave.
- d. Show that the period of the wave is 0,625 s.
- e. Calculate the frequency of the waves.
- f. Calculate the velocity of the waves.

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(1.) 003m (2.) 003n

Longitudinal waves

9

Introduction and key concepts

 ESACR

We have already studied transverse pulses and waves. In this chapter we look at another type of wave called a *longitudinal* wave. In transverse waves, the motion of the particles in the medium was perpendicular to the direction of the wave. In longitudinal waves, the particles in the medium move *parallel* (in the *same* direction as) to the motion of the wave. Examples of transverse waves (discussed in the previous chapter) are water waves. An example of a longitudinal wave is a sound wave. See introductory video: (📺 Video: VPdim at www.everythingscience.co.za)

What is a longitudinal wave?

 ESACS

DEFINITION: Longitudinal waves

A longitudinal wave is a wave where the particles in the medium move parallel to the direction of propagation of the wave.

When we studied transverse waves we looked at two different motions: the motion of the particles of the medium and the motion of the wave itself. We will do the same for longitudinal waves.

The question is how do we construct such a wave?

A longitudinal wave is seen best in a slinky spring. Do the following investigation to find out more about longitudinal waves.

Activity:*Investigating longitudinal waves*

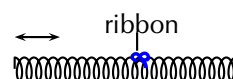
Take a slinky spring and lay it on a table. Hold one end and pull the free end of the spring and flick it back and forth once in the direction of the spring. Observe what happens.

In which direction does the disturbance move?

A slinky spring

Photograph by Tim Ebbs on Flickr.com

flick spring back and forth

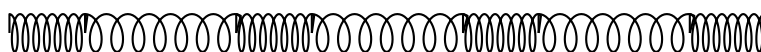


Tie a ribbon to the middle of the spring. Watch carefully what happens to the ribbon when the end of the spring is flicked. Describe the motion of the ribbon.

Flick the spring back and forth continuously to set up a train of pulses, a longitudinal wave.

From the investigation you will have noticed that the disturbance moves parallel to the direction in which the spring was pulled. The ribbon in the investigation represents one particle in the medium. The particles in the medium move in the same direction as the wave. ▶ See video: VPdkf at www.everythingscience.co.za

→ direction of motion of wave



↔ motion of particles in spring is back and forth

Figure 9.2: Longitudinal wave through a spring

As in the case of transverse waves the following properties can be defined for longitudinal waves: wavelength, amplitude, period, frequency and wave speed.

Compression and rarefaction



However instead of crests and troughs, longitudinal waves have *compressions* and *rarefactions*.

DEFINITION: Compression

A **compression** is a region in a longitudinal wave where the particles are closest together.

DEFINITION: Rarefaction

A **rarefaction** is a region in a longitudinal wave where the particles are furthest apart.

▶ See video: VPdml at www.everythingscience.co.za

As seen in Figure 9.3, there are regions where the medium is compressed and other regions where the medium is spread out in a longitudinal wave.

The region where the medium is compressed is known as a **compression** and the region where the medium is spread out is known as a **rarefaction**.

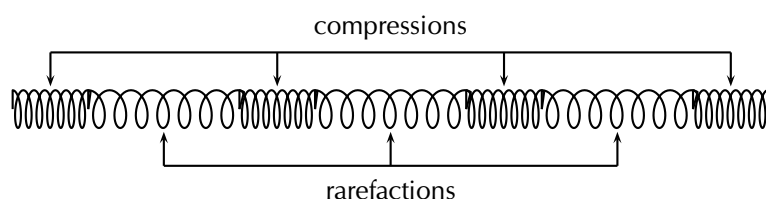


Figure 9.3: Compressions and rarefactions on a longitudinal wave

Wavelength and amplitude

DEFINITION: Wavelength

The **wavelength** in a longitudinal wave is the distance between two consecutive points that are in phase.

The wavelength in a longitudinal wave refers to the distance between two consecutive compressions or between two consecutive rarefactions.

DEFINITION: Amplitude

The **amplitude** is the maximum displacement from equilibrium. For a longitudinal wave which is a pressure wave this would be the maximum increase (or decrease) in pressure from the equilibrium pressure that is caused when a compression (or rarefaction) passes a point.

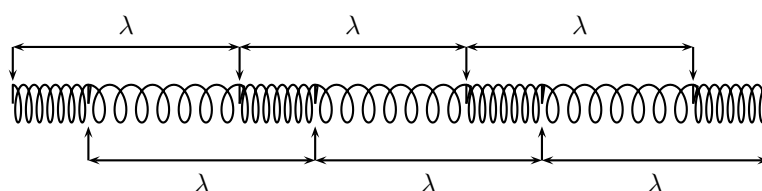


Figure 9.4: Wavelength of a longitudinal wave

The amplitude is the distance from the equilibrium position of the medium to a compression or a rarefaction.

Period and frequency



DEFINITION: Period

The **period** of a wave is the time taken by the wave to move one wavelength.

DEFINITION: Frequency

The **frequency** of a wave is the number of wavelengths per second.

The *period* of a longitudinal wave is the time taken by the wave to move one wavelength. As for transverse waves, the symbol T is used to represent period and period is measured in seconds (s).

The *frequency* f of a wave is the number of wavelengths per second. Using this definition and the fact that the period is the time taken for 1 wavelength, we can define:

$$f = \frac{1}{T}$$

or alternately,

$$T = \frac{1}{f}$$

Speed of a longitudinal wave



The speed of a longitudinal wave is defined in the same way as the speed of transverse waves:

DEFINITION: Wave speed

Wave speed is the distance a wave travels per unit time.

Quantity: Wave speed (v)

Unit name: speed

Unit: $\text{m} \cdot \text{s}^{-1}$

The distance between two successive compressions is 1 wavelength, λ . Thus in a time of 1 period, the wave will travel 1 wavelength in distance. Thus the speed of the wave, v , is:

$$v = \frac{\text{distance travelled}}{\text{time taken}} = \frac{\lambda}{T}$$

However, $f = \frac{1}{T}$. Therefore, we can also write:

$$\begin{aligned}
 v &= \frac{\lambda}{T} \\
 &= \lambda \cdot \frac{1}{T} \\
 &= \lambda \cdot f
 \end{aligned}$$

We call this equation the *wave equation*. To summarise, we have that $v = \lambda \cdot f$ where

- v = speed in $\text{m} \cdot \text{s}^{-1}$
- λ = wavelength in m
- f = frequency in Hz

Example 1: Speed of longitudinal waves

QUESTION

The musical note “A” is a sound wave. The note has a frequency of 440 Hz and a wavelength of 0,784 m. Calculate the speed of the musical note.

SOLUTION

Step 1 : Determine what is given and what is required

Using:

$$\begin{aligned}
 f &= 440 \text{ Hz} \\
 \lambda &= 0,784 \text{ m}
 \end{aligned}$$

We need to calculate the speed of the musical note “A”.

Step 2 : Determine how to approach based on what is given

We are given the frequency and wavelength of the note. We can therefore use:

$$v = f \cdot \lambda$$

Step 3 : Calculate the wave speed

$$\begin{aligned}
 v &= f \cdot \lambda \\
 &= (440 \text{ Hz})(0,784 \text{ m}) \\
 &= 345 \text{ m} \cdot \text{s}^{-1}
 \end{aligned}$$

Step 4 : Write the final answer

The musical note “A” travels at $345 \text{ m} \cdot \text{s}^{-1}$.

Example 2: Speed of longitudinal waves**QUESTION**

A longitudinal wave travels into a medium in which its speed increases. How does this affect its... (write only increases, decreases, stays the same).

1. period?
2. wavelength?

SOLUTION**Step 1 : Determine what is required**

We need to determine how the period and wavelength of a longitudinal wave change when its speed increases.

Step 2 : Determine how to approach based on what is given

We need to find the link between period, wavelength and wave speed.

Step 3 : Discuss how the period changes

We know that the frequency of a longitudinal wave is dependent on the frequency of the vibrations that lead to the creation of the longitudinal wave. Therefore, the frequency is always unchanged, irrespective of any changes in speed. Since the period is the inverse of the frequency, the period remains the same.

Step 4 : Discuss how the wavelength changes

The frequency remains unchanged. According to the wave equation

$$v = f \cdot \lambda$$

if f remains the same and v increases, then λ , the wavelength, must also increase.

Chapter 9 | Summary

See the summary presentation (Presentation: VPduk at www.everythingscience.co.za)

- A longitudinal wave is a wave where the particles in the medium move parallel to the direction in which the wave is travelling.
- Most longitudinal waves consist of areas of higher pressure, where the particles in the medium are closest together (compressions) and areas of lower pressure, where the particles in the medium are furthest apart (rarefactions).
- The wavelength of a longitudinal wave is the distance between two consecutive compressions, or two consecutive rarefactions.
- The relationship between the period (T) and frequency (f) is given by

$$T = \frac{1}{f} \text{ or } f = \frac{1}{T}$$

- The relationship between wave speed (v), frequency (f) and wavelength (λ) is given by

$$v = f\lambda$$

Physical Quantities		
Quantity	Unit name	Unit symbol
Amplitude (A)	metre	m
Wavelength (λ)	metre	m
Period (T)	second	s
Frequency (f)	hertz	Hz (s^{-1})
Wave speed (v)	metre per second	$m \cdot s^{-1}$

Table 9.1: Units used in **longitudinal waves**

Chapter 9

End of chapter exercises

1. Which of the following is not a longitudinal wave?
 - a. light
 - b. sound
 - c. ultrasound
2. Which of the following media can a longitudinal wave like sound not travel through?

- a. solid
 - b. liquid
 - c. gas
 - d. vacuum
3. A longitudinal wave has a compression to compression distance of 10 m. It takes the wave 5 s to pass a point.
- a. What is the wavelength of the longitudinal wave?
 - b. What is the speed of the wave?
4. A flute produces a musical sound travelling at a speed of $320 \text{ m} \cdot \text{s}^{-1}$. The frequency of the note is 256 Hz. Calculate:
- a. the period of the note
 - b. the wavelength of the note

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(1.) 003p (2.) 003q (3.) 003r (4.) 003s

Introduction

 ESACX

Have you ever thought about how amazing your sense of hearing is? It is actually pretty remarkable that we can hear the huge range of sounds and determine direction so quickly. How does something actually make a sound that you can hear? Anything that generates a disturbance in the air creates a pulse that travels away from the place where it was created. If this pulse enters your ear it can cause your ear drum to vibrate which is how you hear. If the source of the pulse creates a train of pulses then the disturbance is a wave. See introductory video: (▶) Video: VPdxu at www.everythingscience.co.za

We generally say that sound is a wave. Sound waves are longitudinal, pressure waves, that means that the waves consist of compressions and rarefactions of the pressure of the air.

Sound waves

 ESACY

A tuning fork is an instrument used by musicians to create sound waves of a specific frequency. They are often used to tune musical instruments.

Sound waves coming from a tuning fork are caused by the vibrations of the tuning fork which push against the air particles in front of it. As the air particles are pushed together a compression is formed. The particles behind the compression move further apart causing a rarefaction. As the particles continue to push against each other, the sound wave travels through the air. Due to this motion of the particles, there is a constant variation in the pressure in the air. Sound waves are therefore pressure waves. This means that in media where the particles are closer together, sound waves will travel faster.

▶ See video: VPdya at www.everythingscience.co.za

Tuning fork



Photo by amonya on Flickr.com

Sound waves travel faster through liquids, like water, than through the air because water is denser than air (the particles are closer together). Sound waves travel faster in solids than in liquids.

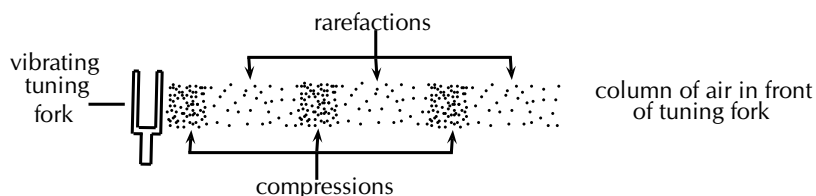


Figure 10.2: Sound waves are pressure waves and need a medium through which to travel.

Activity:

Build your own telephone

Have you ever wondered if you can actually use tin cans or cups to make a telephone? Try it and see!

What you need:

- Two tin cans or paper cups
- String
- Toothpicks or small sticks

Try This:

1. Tie a toothpick on each end of a length of string.
2. Make a hole in the base of the can or cup. Poke the toothpick through the end of the can. Pull the string tight so the toothpick rests on the inside bottom of the can. Put one can at each end of the string. (You may want to experiment with different cans or cups and strings or wires to see what works best.)
3. Hold the string tight and talk into one of the cans. The person at the other end should be able to hear you. Why does the string have to be tight?
4. Try to make a party line by tying a third string and can or cup onto the middle of the string. Can everybody talk to everybody else?

Sound waves need something to travel through. Usually they travel through air, but they can travel much faster and farther through a string. The string has to be tight or else the sound wave cannot travel through it. The cup helps to amplify the sound on the other end.

Tip

A sound wave is a pressure wave. This means that regions of high pressure (compressions) and low pressure (rarefactions) are created as the sound source vibrates. These compressions and rarefactions arise because the source vibrates longitudinally and the longitudinal motion of air produces pressure fluctuations.

Speed of sound



FACT

The speed of sound in air, at sea level, at a temperature of 21°C and under normal atmospheric conditions, is $344 \text{ m} \cdot \text{s}^{-1}$.

The speed of sound depends on the medium the sound is travelling in. Sound travels faster in solids than in liquids, and faster in liquids than in gases. This is because the density of solids is higher than that of liquids which means that the particles are closer together. Sound can be transmitted more easily.

▶ See video: VPeac at www.everythingscience.co.za

The speed of sound also depends on the temperature of the medium. The hotter the medium is, the faster its particles move and therefore the quicker the sound will travel through the medium. When we heat a substance, the particles in that substance have more kinetic energy and vibrate or move faster. Sound can therefore be transmitted more easily and quickly in hotter substances.

Sound waves are pressure waves. The speed of sound will therefore be influenced by the pressure of the medium through which it is travelling. At sea level the air pressure is higher than high up on a mountain. Sound will travel faster at sea level where the air pressure is higher than it would at places high above sea level.

Substance	$v \text{ (m} \cdot \text{s}^{-1}\text{)}$
aluminium	6420
brick	3650
copper	4760
glass	5100
gold	3240
lead	2160
water, sea	1531
air, 0°C	331
air, 20°C	343

Table 10.1: The speed of sound in different materials.

Informal experiment: *Measuring the speed of sound in air*

Aim: To measure the speed of sound.

Apparatus:

- Starter's gun or anything that can produce a loud sound in response to visible action

- Stopwatch

Method: The speed of sound can be measured because light travels much faster than sound. Light travels at about $300\,000\text{ m} \cdot \text{s}^{-1}$ (you will learn more about the speed of light in the next chapter) while sound only travels at about $300\text{ m} \cdot \text{s}^{-1}$. This difference means that over a distance of 300 m, the light from an event will reach your eyes almost instantly but there will be an approximate half a second lag before you hear the sound produced. Thus if a starter's pistol is fired from a great distance, you will see the smoke immediately but there will be a lag before you hear the sound. If you know the distance and the time then you can calculate the speed (distance divided by time). You don't need a gun but anything that you can see producing a loud sound.

Try this:

1. Find a place where you know the precise, straight-line distance between two points (maybe an athletics track)
2. Someone needs to stand at the one point to produce the sound
3. Another person needs to stand at the other point with the stop watches
4. The person with the stopwatch should start the stopwatch when they see the other person make the sound and stop the stopwatch when they hear the sound (do this a few times and write the times down)

Results:

You can now calculate the speed to sound by dividing the distance by the time. Remember to work in S.I. units (metres and seconds). If you took multiple readings then you can sum them and divide by the number of readings to get an average time reading. Use the average time to calculate the speed:

$$v = \frac{D}{t}$$

Time (s)	Distance (m)	$v \text{ (m} \cdot \text{s}^{-1}\text{)}$
Averages		

Conclusions: Some questions to ask:

- What is your reaction time on the stopwatch? You can test this by starting it and then trying to stop it immediately.
- What was the forecast temperature on the day of the measurement?
- Was it humid or very dry?

Discuss what might change the speed of sound that you measured.

You can vary this experiment by trying it on days when the weather is different as this can change air pressure and temperature.

Reflection and echoes



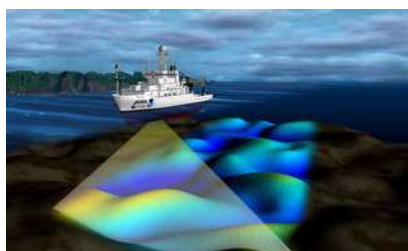
When the sound waves collide with an object they are reflected. You can think of the individual particles that are oscillating about their equilibrium position colliding into the object when the wave passes. They bounce off the object causing the wave to be reflected.

In a space with many small objects there are reflections at every surface but they are too small and too mixed up to have an outcome that a human can hear. However, when there is an open space that has only large surfaces, for example a school hall that is empty, then the reflected sound can actually be heard. The sound wave is reflected in such a wave that the wave looks the same but is moving in the opposite direction. ▶ See video: VPebb at www.everythingscience.co.za

This means that if you stand in a hall and loudly say “hello” you will hear yourself say “hello” a split second later. This is an echo. This can also happen outdoors in a wide open space with a large reflecting surface nearby, like standing near a mountain cliff in an area with no trees or bushes.

This is a very useful property of waves.

SONAR



Ships on the ocean make use of the reflecting properties of sound waves to determine the depth of the ocean. A sound wave is transmitted and bounces off the seabed. Because the speed of sound is known and the time lapse between sending and receiving the sound can be measured, the distance from the ship to the bottom of the ocean can be determined. This is called sonar, which is an acronym for **S**ound **N**avigation **A**nd **R**anging.

Example 1: SONAR**QUESTION**

A ship sends a signal to the bottom of the ocean to determine the depth of the ocean. The speed of sound in sea water is $1450 \text{ m} \cdot \text{s}^{-1}$. If the signal is received 1,5 seconds later, how deep is the ocean at that point?

SOLUTION

Step 1 : **Identify what is given and what is being asked:**

$$\begin{aligned}s &= 1450 \text{ m} \cdot \text{s}^{-1} \\ t &= 1,5 \text{ s there and back} \\ \therefore t &= 0,75 \text{ s one way} \\ d &= ?\end{aligned}$$

Step 2 : **Calculate the distance:**

$$\begin{aligned}\text{Distance} &= \text{speed} \times \text{time} \\ D &= s \times t \\ &= 1450 \text{ m} \cdot \text{s}^{-1} \times 0,75 \text{ s} \\ &= 1087,5 \text{ m}\end{aligned}$$

Echolocation



Animals like dolphins and bats make use of sound waves to find their way. Just like ships on the ocean, bats use sonar to navigate. Waves that are sent out are reflected off the objects around the animal. Bats, or dolphins, then use the reflected sounds to form a “picture” of their surroundings. This is called echolocation.

Characteristics of a sound wave



Since sound is a wave, we can relate the properties of sound to the properties of a wave. The basic properties of sound are: pitch, loudness and tone.

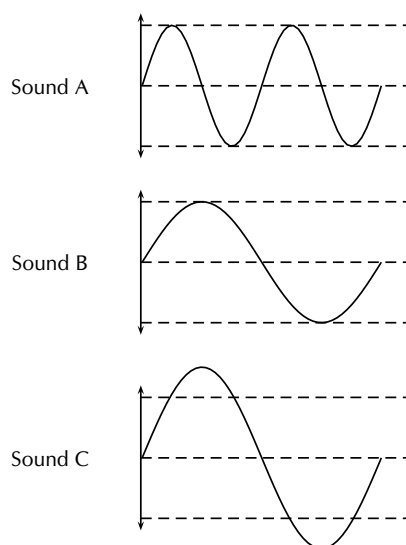


Figure 10.3: Pitch and loudness of sound. Sound B has a *lower* pitch (lower frequency) than Sound A and is *softer* (smaller amplitude) than Sound C.

Pitch

The frequency of a sound wave is what your ear understands as pitch. A higher frequency sound has a higher pitch, and a lower frequency sound has a lower pitch. In Figure 10.3 sound A has a higher pitch than sound B. For instance, the chirp of a bird would have a high pitch, but the roar of a lion would have a low pitch.

The human ear can detect a wide range of frequencies. Frequencies from 20 to 20 000 Hz are audible to the human ear. Any sound with a frequency below 20 Hz is known as an **infrasound** and any sound with a frequency above 20 000 Hz is known as an **ultrasound**.

Table 10.2 lists the ranges of some common animals compared to humans.

Table 10.2: Range of frequencies

	lower frequency (Hz)	upper frequency (Hz)
Humans	20	20 000
Dogs	50	45 000
Cats	45	85 000
Bats	20	120 000
Dolphins	0,25	200 000
Elephants	5	10 000

Activity:*Range of wavelengths*

Using the information given in Table 10.2, calculate the lower and upper wavelengths that each species can hear. Assume the speed of sound in air is $344 \text{ m} \cdot \text{s}^{-1}$.

Loudness

The amplitude of a sound wave determines its loudness or volume. A larger amplitude means a louder sound, and a smaller amplitude means a softer sound. In Figure 10.3 sound C is louder than sound B. The vibration of a source sets the amplitude of a wave. It transmits energy into the medium through its vibration. More energetic vibration corresponds to larger amplitude. The molecules move back and forth more vigorously.

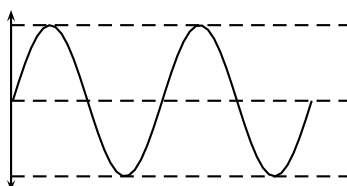
The loudness of a sound is also determined by the sensitivity of the ear. The human ear is more sensitive to some frequencies than to others. The volume we receive thus depends on both the amplitude of a sound wave and whether its frequency lies in a region where the ear is more or less sensitive.

Exercise 10 - 1

Study the following diagram representing a musical note. Redraw the diagram for a note

1. with a higher pitch
2. that is louder

3. that is softer



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(1.) 003t

Activity:

Comparing sound generating instruments

The size and shape of instruments influences the sounds that they are able to produce. Find some instruments that have different physical characteristics and compare their sounds. You could:

Option 1: Vuvuzelas:

Compare the sounds made by blowing through vuvuzelas of different sizes. You will need to find a few different vuvuzelas. Take turns blowing the different ones, one at a time and record which you think is louder (amplitude), which is of higher pitch (frequency).

Option 2: Tuning forks:

Compare the sounds created by tapping tuning forks of different sizes. You will need to find a few different tuning forks. Take turns tapping the different ones, one at a time and record which you think is louder (amplitude), which is of higher pitch (frequency).

Option 3: Signal generator and oscilloscope:

Use a function generator connected to a speaker to produce sounds of different frequencies and amplitudes and use a microphone connected to an oscilloscope to display the characteristics of the different sounds produced.

Function generator

The function generator allows you to control the loudness and frequency of the sound being produced by the speaker. It will have controls for amplitude and frequency.

A function generator



Photograph by Audin on Flickr.com

Oscilloscope

The microphone can then pick up the sound and convert it to an electrical signal which can be displayed on the oscilloscope.

The most common oscilloscope controls are for amplitude, frequency, triggering, and channels. Once your teacher has helped you acquire a signal using the correct channel and triggering you will use the amplitude and frequency controls to display the characteristics of the sound being produced.

The amplitude adjustment of an oscilloscope controls how tall a given voltage will appear on the screen. The purpose of this adjustment is that you can see a very large or a very small signal on the same screen.

The frequency (or time) adjustment of an oscilloscope is how much time will a certain distance across the screen represent. The purpose of this adjustment is to be able to see a very quickly changing or a slowly changing signal on the same screen.

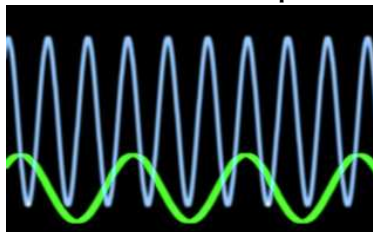
You will be able to experiment with different amplitudes and frequencies using the function generator and see what impact the changes have on the waveform picked up by the microphone.

An oscilloscope



Photograph by Audin on Flickr.com

Two different oscilloscope traces



Photograph by Audin on Flickr.com

Note: The display of the oscilloscope will show you a transverse wave pattern. This does not mean that sound waves are transverse waves but just shows that the pressure being measured is fluctuating because of a pressure wave.

▶ See video: VPebn at www.everythingscience.co.za

Intensity of sound [NOT IN CAPS]



Intensity is one indicator of amplitude. Intensity is the energy transmitted over a unit of area each second.

The unit of intensity is the **decibel** (symbol: dB).

Source	Intensity (dB)	Times greater than hearing threshold
Rocket Launch	180	10^{18}
Jet Plane	140	10^{14}
Threshold of Pain	120	10^{12}
Rock Band	110	10^{11}
Factory	80	10^8
City Traffic	70	10^7
Normal Conversation	60	10^6
Library	40	10^4
Whisper	20	10^2
Threshold of hearing	0	0

Table 10.3: Examples of sound intensities.

Vuvuzelas feature prominently at soccer events in South Africa. The intensity of the sound from a vuvuzela depends on how close you are. In Table 10.3 you can see how the intensity differs.

Vuvuzelas in action



Photo by Dundas Football Club on Flickr.com

Frequency (Hz)	At ear	Bell opening	1 m	2 m
125	36	62	38	35
250	92	106	82	85
500	103	121	102	101
1 000	106	122	108	100
2 000	101	122	110	101
4 000	97	109	110	102
5 000	93	111	109	100
8 000	87	110	107	98

Table 10.4: Average vuvuzela intensity measurements across frequencies at 4 distinct distances from the bell end of the vuvuzela (dBA) taken from South African Medical Journal (Cape Town, South Africa) **100** (4): 192

Ultrasound



Ultrasound is sound with a frequency that is higher than 20 kHz. Some animals, such as dogs, dolphins, and bats, have an upper limit that is greater than that of the human ear and can hear ultrasound.

Application	Lowest Frequency (kHz)	Highest Frequency (kHz)
Cleaning (e.g. jewellery)	20	40
Material testing for flaws	50	500
Welding of plastics	15	40
Tumour ablation	250	2000

Table 10.5: Different uses of ultrasound and the frequencies applicable.

Ultrasound image of a unborn baby

Picture by mbaylor on Flickr.com

FACT

Ultrasound generator-/speaker systems are sold with claims that they frighten away rodents and insects, but there is no scientific evidence that the devices work; controlled tests have shown that rodents quickly learn that the speakers are harmless.

The most common use of ultrasound is to create images, and has industrial and medical applications. The use of ultrasound to create images is based on the reflection and transmission of a wave at a boundary (when the wave goes from one substance to another). When an ultrasound wave travels inside an object that is made up of different materials such as the human body, each time it encounters a boundary, e.g. between bone and muscle, or muscle and fat, part of the wave is reflected and part of it is transmitted. The reflected rays are detected and used to construct an image of the object.

Ultrasound in medicine can visualise muscle and soft tissue, making them useful for scanning the organs, and is commonly used during pregnancy. Ultrasound is a safe, non-invasive method of looking inside the human body.

Ultrasound sources may be used to generate local heating in biological tissue, with applications in physical therapy and cancer treatment. Focused ultrasound sources may be used to break up kidney stones.

Ultrasonic cleaners, sometimes called supersonic cleaners, are used at frequencies from 20-40 kHz for jewellery, lenses and other optical parts, watches, dental instruments, surgical instruments and industrial parts. These cleaners consist of containers with a fluid in which the object to be cleaned is placed. Ultrasonic waves are then sent into the fluid. The main mechanism for cleaning action in an ultrasonic cleaner is actually the energy released from the collapse of millions of microscopic bubbles occurring in the liquid of the cleaner.

The physics of hearing **[NOT IN CAPS]**

www.ESADG

The human ear is divided into three main sections: the outer, middle, and inner ear. Let's follow the journey of a sound wave from the pinna (outermost part) to the auditory nerve (innermost part) which transmits a signal to the brain. The pinna is the part of the ear we typically think of when we refer to the ear. Its main function is to collect and focus a sound wave. The wave then travels through the ear canal until it meets the eardrum. The pressure fluctuations of the sound wave make the eardrum vibrate. The three very small bones of the middle ear, the malleus (hammer), the incus (anvil), and the stapes (stirrup), transmit the

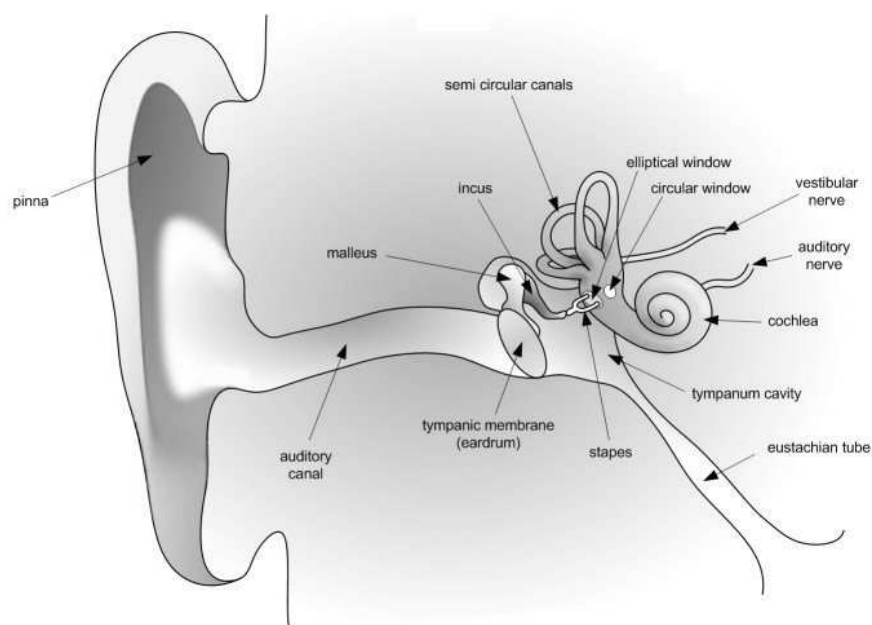


Figure 10.4: Diagram of the human ear.

signal through to the elliptical window. The elliptical window is the beginning of the inner ear. From the elliptical window the sound waves are transmitted through the liquid in the inner ear and interpreted as sounds by the brain. The inner ear, made of the semicircular canals, the cochlea, and the auditory nerve, is filled with fluid. The fluid allows the body to detect quick movements and maintain balance.

There are sounds which exceed the threshold of pain. Exposure to these sounds can cause immediate damage to hearing. In fact, exposure to sounds from 80 dB and above can damage hearing over time. Measures can be taken to avoid damage, such as wearing earplugs or ear muffs. Limiting exposure time and increasing distance between you and the source are also important steps for protecting your hearing.

Group Discussion: Importance of Safety Equipment

Working in groups of 5, discuss the importance of safety equipment such as ear protectors for workers in loud environments, e.g. those who use jack hammers or direct aeroplanes to their parking bays. Write up your conclusions in a one page report. Some prior research into the importance of safety equipment might be necessary to complete this group discussion.

Chapter 10 | Summary

See the summary presentation (📺 Presentation: VPecu at www.everythingscience.co.za)

- Sound waves are longitudinal waves
- The **frequency** of a sound is an indication of how high or low the *pitch* of the sound is.
- The human ear can hear frequencies from 20 to 20 000 Hz. **Infrasound** waves have frequencies lower than 20 Hz. **Ultrasound** waves have frequencies higher than 20 000 Hz.
- The **amplitude** of a sound determines its *loudness* or volume.
- The **tone** is a measure of the *quality* of a sound wave.
- The speed of sound in air is around $340 \text{ m} \cdot \text{s}^{-1}$. It is dependent on the temperature, height above sea level and the phase of the medium through which it is travelling.
- Sound travels faster when the medium is hot.
- Sound travels faster in a solid than a liquid and faster in a liquid than in a gas.
- Sound travels faster at sea level where the air pressure is higher.
- The intensity of a sound is the energy transmitted over a certain area. Intensity is a measure of frequency.
- Ultrasound can be used to form pictures of things we cannot see, like unborn babies or tumours.
- Echolocation is used by animals such as dolphins and bats to “see” their surroundings by using ultrasound.
- Ships use sonar to determine how deep the ocean is or to locate shoals of fish.

Physical Quantities		
Quantity	Unit name	Unit symbol
Velocity (v)	metre per second	$\text{m} \cdot \text{s}^{-1}$
Wavelength (λ)	metre	m
Amplitude (A)	metre	m
Period (T)	second	s
Frequency (f)	hertz	Hz (s^{-1})

Table 10.6: Units used in **sound**

Chapter 10

End of chapter exercises

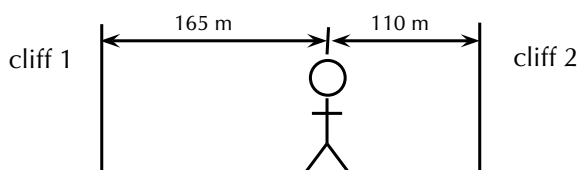
1. Choose a word from column B that best describes the concept in column A.

Column A	Column B
1. pitch of sound	A. amplitude
2. loudness of sound	B. frequency
3. quality of sound	C. speed
	D. waveform

2. A tuning fork, a violin string and a loudspeaker are producing sounds. This is because they are all in a state of:
- compression
 - rarefaction
 - rotation
 - tension
 - vibration
3. What would a drummer do to make the sound of a drum give a note of lower pitch?
- hit the drum harder
 - hit the drum less hard
 - hit the drum near the edge
 - loosen the drum skin
 - tighten the drum skin
4. What is the approximate range of audible frequencies for a healthy human?
- 0.2 Hz → 200 Hz
 - 2 Hz → 2 000 Hz
 - 20 Hz → 20 000 Hz
 - 200 Hz → 200 000 Hz
 - 2 000 Hz → 2 000 000 Hz
5. X and Y are different wave motions. In air, X travels much faster than Y but has a much shorter wavelength. Which types of wave motion could X and Y be?

	<u>X</u>	<u>Y</u>
1.	microwaves	red light
2.	radio	infra red
3.	red light	sound
4.	sound	ultraviolet
5.	ultraviolet	radio

6. Astronauts are in a spaceship orbiting the moon. They see an explosion on the surface of the moon. Why can they not hear the explosion?
- explosions do not occur in space
 - sound cannot travel through a vacuum
 - sound is reflected away from the spaceship
 - sound travels too quickly in space to affect the ear drum
 - the spaceship would be moving at a supersonic speed
7. A man stands between two cliffs as shown in the diagram and claps his hands once.



Assuming that the velocity of sound is $330 \text{ m} \cdot \text{s}^{-1}$, what will be the time interval between the two loudest echoes?

- $\frac{2}{3} \text{ s}$
 - $\frac{1}{6} \text{ s}$
 - $\frac{5}{6} \text{ s}$
 - 1 s
 - $\frac{1}{3} \text{ s}$
8. A dolphin emits an ultrasonic wave with frequency of 0,15 MHz. The speed of the ultrasonic wave in water is $1500 \text{ m} \cdot \text{s}^{-1}$. What is the wavelength of this wave in water?
- 0,1 mm
 - 1 cm
 - 10 cm
 - 10 m
 - 100 m
9. The amplitude and frequency of a sound wave are both increased. How are the loudness and pitch of the sound affected?

	<u>loudness</u>	<u>pitch</u>
A	increased	raised
B	increased	unchanged
C	increased	lowered
D	decreased	raised
E	decreased	lowered

10. A jet fighter travels slower than the speed of sound. Its speed is said to be:

- a. Mach 1
- b. supersonic
- c. subsonic
- d. hypersonic
- e. infrasonic

11. A sound wave is different from a light wave in that a sound wave is:

- a. produced by a vibrating object and a light wave is not.
- b. not capable of travelling through a vacuum.
- c. not capable of diffracting and a light wave is.
- d. capable of existing with a variety of frequencies and a light wave has a single frequency.

12. At the same temperature, sound waves have the fastest speed in:

- a. rock
- b. milk
- c. oxygen
- d. sand

13. Two sound waves are travelling through a container of nitrogen gas. The first wave has a wavelength of 1,5 m, while the second wave has a wavelength of 4,5 m. The velocity of the second wave must be:

- a. $\frac{1}{9}$ the velocity of the first wave.
- b. $\frac{1}{3}$ the velocity of the first wave.
- c. the same as the velocity of the first wave.
- d. three times larger than the velocity of the first wave.
- e. nine times larger than the velocity of the first wave.

14. A lightning storm creates both lightning and thunder. You see the lightning almost immediately since light travels at $3 \times 10^8 \text{ m} \cdot \text{s}^{-1}$. After seeing the lightning, you count 5 s and then you hear the thunder. Calculate the distance to the location of the storm.

15. A person is yelling from a second story window to another person standing at the garden gate, 50 m away. If the speed of sound is $344 \text{ m} \cdot \text{s}^{-1}$,

how long does it take the sound to reach the person standing at the gate?

16. Person 1 speaks to person 2. Explain how the sound is created by person 1 and how it is possible for person 2 to hear the conversation.
17. Sound cannot travel in space. Discuss what other modes of communication astronauts can use when they are outside the space shuttle?
18. An automatic focus camera uses an ultrasonic sound wave to focus on objects. The camera sends out sound waves which are reflected off distant objects and return to the camera. A sensor detects the time it takes for the waves to return and then determines the distance an object is from the camera. If a sound wave (speed = $344 \text{ m}\cdot\text{s}^{-1}$) returns to the camera 0,150 s after leaving the camera, how far away is the object?
19. Calculate the frequency (in Hz) and wavelength of the annoying sound made by a mosquito when it beats its wings at the average rate of 600 wing beats per second. Assume the speed of the sound waves is $344 \text{ m}\cdot\text{s}^{-1}$.
20. How does halving the frequency of a wave source affect the speed of the waves?
21. Humans can detect frequencies as high as 20 000 Hz. Assuming the speed of sound in air is $344 \text{ m}\cdot\text{s}^{-1}$, calculate the wavelength of the sound corresponding to the upper range of audible hearing.
22. An elephant trumpets at 10 Hz. Assuming the speed of sound in air is $344 \text{ m}\cdot\text{s}^{-1}$, calculate the wavelength of this infrasonic sound wave made by the elephant.
23. A ship sends a signal out to determine the depth of the ocean. The signal returns 2,5 seconds later. If sound travels at $1450 \text{ m}\cdot\text{s}^{-1}$ in sea water, how deep is the ocean at that point?
24. A person shouts at a cliff and hears an echo from the cliff 1 s later. If the speed of sound is $344 \text{ m}\cdot\text{s}^{-1}$, how far away is the cliff?
25. Select a word from Column B that best fits the description in Column A:

Column A	Column B
1. waves in the air caused by vibrations	A. longitudinal waves
2. waves that move in one direction, but medium moves in another	B. frequency
3. waves and medium that move in the same direction	C. period
4. the distance between consecutive points of a wave which are in phase	D. amplitude
5. how often a single wavelength goes by	E. sound waves
6. half the difference between high points and low points of waves	F. standing waves
7. the distance a wave covers per time interval	G. transverse waves
8. the time taken for one wavelength to pass a point	H. wavelength
	I. music
	J. sounds
	K. wave speed

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- (1.) 003u (2.) 003v (3.) 003w (4.) 003x (5.) 003y (6.) 003z
 (7.) 0040 (8.) 0041 (9.) 0042 (10.) 0043 (11.) 0044 (12.) 0045
 (13.) 0046 (14.) 0048 (15.) 0049 (16.) 004c (17.) 004d (18.) 004e
 (19.) 004f (20.) 004g (21.) 004h (22.) 004i (23.) 004j (24.) 004k
 (25.) 004m

What is electromagnetic radiation?



The most common example of electromagnetic (EM) radiation is visible light. Everyone is very familiar with light in everyday life, you can only see things because light bounces off them and enters your eyes. This alone makes it worthwhile to learn about it but there are also very many other applications of EM radiation. It is called electromagnetic because there are electric and magnetic fields making up the radiation. We will look at this in more detail a little later. See introductory video: (▶ Video: VPefg at www.everythingscience.co.za)

In everyday experience, light doesn't seem to have many special properties but it does:

- **A huge spectrum:** The light we can see (visible EM radiation) is only a small part of all of the EM radiation (electromagnetic spectrum) that exists.
- **Nature's speed limit:** Nothing moves faster than the speed of light.
- **Wave nature:** All EM radiation has the ability to behave like a wave which we call wave-like behaviour.
- **Particle nature:** All EM radiation has the ability to behave like a particle which we call particle-like behaviour.
- **No medium required:** EM radiation can propagate without a medium through which to move even though they are waves.

Two important things to notice, we have mentioned:

1. waves without a medium and
2. being both a particle and a wave.

We will discuss this in the following sections and in even more detail in Grades 11 and 12.

Wave-like nature of EM radiation



If you watch a colony of ants walking up the wall, they look like a thin continuous black line. But as you look closer, you see that the line is made up of thousands of separated black ants.

Light and all other types of electromagnetic radiation seem like a continuous wave at first, but when one performs experiments with light, one can notice that light can have both wave- and particle-like properties. Just like the individual ants, light can also be made up of individual bundles of energy, or quanta of light.

Light has both wave-like and particle-like properties (wave–particle duality), but only shows one or the other, depending on the kind of experiment we perform. A wave-type experiment shows the wave nature, and a particle-type experiment shows particle nature. One **cannot** test the wave and the particle nature at the same time. A particle of light is called a photon.

DEFINITION: Photon

A photon is a quantum (energy packet) of light.

Fields



Accelerating charges emit electromagnetic waves. A changing electric field generates a magnetic field and a changing magnetic field generates an electric field. This is the principle behind the propagation of electromagnetic waves, because electromagnetic waves, unlike sound waves, do not need a medium to travel through. ▶ See video: VPeka at www.everythingscience.co.za

EM waves propagate when an electric field oscillating in one plane produces a magnetic field oscillating in a plane at right angles to it, which produces an oscillating electric field, and so on. The propagation of electromagnetic waves can be described as *mutual induction*. The changing electric field is responsible for inducing the magnetic field and vice versa.

We use E to denote electric fields and B to denote magnetic fields.

These concepts will be covered in more detail in Grade 11.

These mutually regenerating fields, most commonly described as self-propagating, travel through empty space at a constant speed of approximately $3 \times 10^8 \text{ m} \cdot \text{s}^{-1}$, represented by c . We will use $3 \times 10^8 \text{ m} \cdot \text{s}^{-1}$ for all of our calculations in this book. Although an electromagnetic wave can travel through empty space, it can also travel through a medium (such as water and air). When an electromagnetic wave travels through a medium, it travels slower than it would through empty space (vacuum).

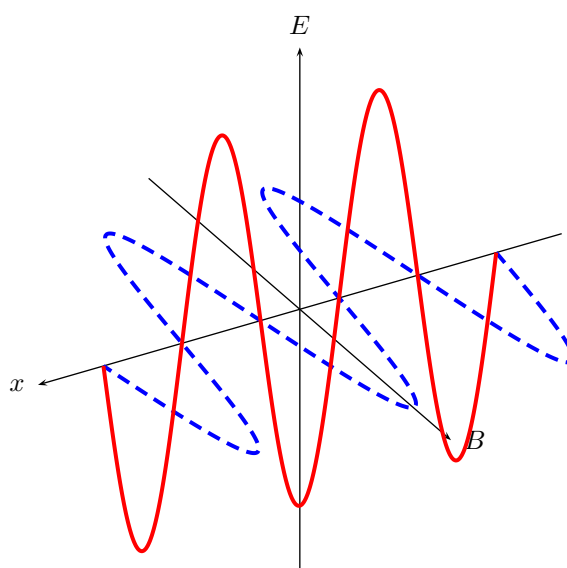


Figure 11.2: A diagram showing the mutually regenerating electric field (red (solid) line) and magnetic field (blue (dashed) line).

Since an EM radiation is a wave, the following equation still applies:

$$v = f \cdot \lambda$$

Except that we can replace v with c :

$$c = f \cdot \lambda$$

Example 1: EM radiation I**QUESTION**

Calculate the frequency of an electromagnetic wave with a wavelength of $4,2 \times 10^{-7} \text{ m}$.

SOLUTION**Step 1 : Wave equation**

We use the formula: $c = f\lambda$ to calculate frequency. The speed of light is a constant $3 \times 10^8 \text{ m} \cdot \text{s}^{-1}$.

Step 2 : Calculate

$$\begin{aligned}c &= f\lambda \\3 \times 10^8 \text{ m} \cdot \text{s}^{-1} &= f \times 4,2 \times 10^{-7} \text{ m} \\f &= 7,14 \times 10^{14} \text{ Hz}\end{aligned}$$

Step 3 : Quote the final answer

The frequency is $7,14 \times 10^{14} \text{ Hz}$.

Example 2: EM Radiation II**QUESTION**

An electromagnetic wave has a wavelength of 200 nm. What is the frequency of the radiation?

SOLUTION**Step 1 : What do we know?**

Recall that all radiation travels at the speed of light (c) in vacuum. Since the question does not specify through what type of material the wave is

travelling, one can assume that it is travelling through a vacuum. We can identify two properties of the radiation - *wavelength* (200 nm) and speed (c).

Step 2 : Apply the wave equation

$$\begin{aligned}c &= f\lambda \\3 \times 10^8 \text{ m} \cdot \text{s}^{-1} &= f \times 200 \times 10^{-9} \text{ m} \\f &= 1.5 \times 10^{15} \text{ Hz}\end{aligned}$$

Step 3 : Quote the final answer

The frequency is 1.5×10^{15} Hz.

Electromagnetic spectrum



EM radiation is classified into types according to the frequency of the wave: these types include, in order of increasing frequency, radio waves, microwaves, infrared radiation, visible light, ultraviolet radiation, X-rays and gamma rays.

► See video: VPemp at www.everythingscience.co.za

Table 11.3 lists the wavelength and frequency ranges of the divisions of the electromagnetic spectrum.

Category	Range of Wavelengths (nm)	Range of Frequencies (Hz)
gamma rays	< 1	$> 3 \times 10^{19}$
X-rays	1-10	3×10^{17} - 3×10^{19}
ultraviolet light	10-400	$7,5 \times 10^{14}$ - 3×10^{17}
visible light	400-700	$4,3 \times 10^{14}$ - $7,5 \times 10^{14}$
infrared	700- 10^5	3×10^{12} - $4,3 \times 10^{19}$
microwave	$10^5 - 10^8$	3×10^9 - 3×10^{12}
radio waves	$> 10^8$	$< 3 \times 10^9$

Table 11.1: Electromagnetic spectrum

Examples of some uses of electromagnetic waves are shown in Table 11.3.

Category	Uses
gamma rays	used to kill the bacteria in marshmallows and to sterilise medical equipment
X-rays	used to image bone structures
ultraviolet light	bees can see into the ultraviolet because flowers stand out more clearly at this frequency
visible light	used by humans to observe the world
infrared	night vision, heat sensors, laser metal cutting
microwave	microwave ovens, radar
radio waves	radio, television broadcasts

Table 11.2: Uses of EM waves

Exercise 11 - 1

1. Arrange the following types of EM radiation in order of increasing frequency: infrared, X-rays, ultraviolet, visible, gamma.

2. Calculate the frequency of an EM wave with a wavelength of 400 nm.
3. Give an example of the use of each type of EM radiation, i.e. gamma rays, X-rays, ultraviolet light, visible light, infrared, microwave and radio and TV waves.

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(1.) 004q (2.) 004r (3.) 004s

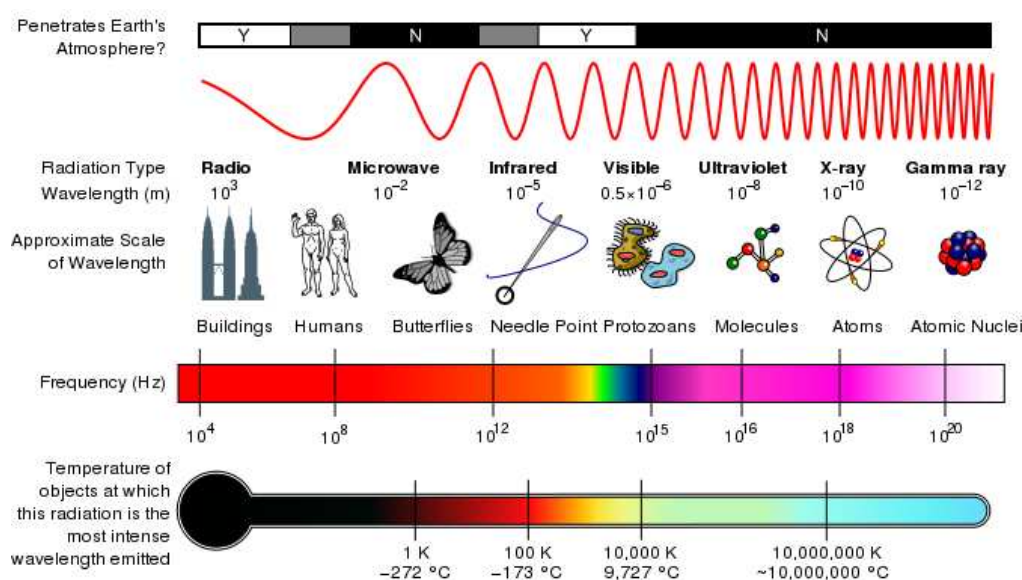


Figure 11.3: The electromagnetic spectrum as a function of frequency. The different types according to wavelength are shown as well as everyday comparisons.

EM radiation in the visible part of the spectrum is scattered off all of the objects around us. This EM radiation provides the information to our eyes that allows us to see. The frequencies of radiation the human eye is sensitive to constitute only a very small part of all possible frequencies of EM radiation. The full set of EM radiation is called the electromagnetic spectrum. To simplify things the EM spectrum divided into sections (such as radio, microwave, infrared, visible, ultraviolet, X-rays and gamma-rays).  See video: VPfdy at www.everythingscience.co.za

The EM spectrum is continuous (has no gaps) and infinite. Due to technological limitations,

we can only use electromagnetic radiation with wavelengths between 10^{-14}m and 10^{15}m .

Penetrating ability of EM radiation



Different frequencies of EM radiation have different degrees of penetration. For example, if we take the human body as the object, visible light is reflected off the surface of the human body, ultra-violet light (from sunlight) damages the skin, but X-rays are able to penetrate the skin and bone and allow for pictures of the inside of the human body to be taken.

If we compare the energy of visible light to the energy of X-rays, we find that X-rays have a much higher frequency. Usually, electromagnetic radiation with higher frequency (energy) have a higher degree of penetration than those with low frequency.

Certain kinds of electromagnetic radiation such as ultra-violet radiation, X-rays and gamma rays are very dangerous. Radiation such as these are called ionising radiation. Ionising radiation transfers energy as it passes through matter, breaking molecular bonds and creating ions.

Excessive exposure to radiation, including sunlight, X-rays and all nuclear radiation, can cause destruction of biological tissue. Luckily, the Earth's atmosphere protects us and other living beings on Earth from most of the harmful EM radiation.

Ultraviolet (UV) radiation and the skin

UVA and UVB are different ranges of frequencies for ultraviolet (UV) light. UVA and UVB can damage collagen fibres which results in the speeding up skin ageing. In general, UVA is the least harmful, but it can contribute to the ageing of skin, DNA damage and possibly skin cancer. It penetrates deeply and does not cause sunburn.

UVB light can cause skin cancer. The radiation excites DNA molecules in skin cells, resulting in possible cancerous mutations. In particular, the layer of ozone in the atmosphere protects us from UVB radiation. The connection between UVB radiation and cancer is one of the reasons for concern about the depletion of ozone in the atmosphere. ▶ See video: VPfeu at www.everythingscience.co.za

As a defence against UV radiation, the body tans when exposed to moderate (depending on skin type) levels of radiation by releasing the brown pigment melanin. This helps to block UV penetration and prevent damage to the vulnerable skin tissue deeper down. Sun-tan lotion, often referred to as sunblock or sunscreen, partly blocks UV radiation and is widely available. These products have a sun protection factor (SPF) rating (usually indicated on the container) that indicate how much protection the product provides against UVB radiation.

The SPF rating does not specify protection against UVA radiation. Some sunscreen lotion now includes compounds such as titanium dioxide which helps protect against UVA rays. Other UVA-blocking compounds found in sunscreen include zinc oxide and avobenzone.

What makes a good sunscreen?

- UVB protection: Padimate O, Homosalate, Octisalate (octyl salicylate), Octinoxate (octyl methoxycinnamate)
- UVA protection: Avobenzone
- UVA/UVB protection: Octocrylene, titanium dioxide, zinc oxide, Mexoryl (ecamsule)



Another means to block UV is by wearing sun protective clothing. This is clothing that has a UPF rating that describes the protection given against both UVA and UVB.

Ultraviolet radiation and the eyes

High intensity UVB light can cause damage to the eyes and exposure can cause welder's flash (photo keratitis or arc eye) and may lead to cataracts and other medical issues.

Protective eyewear is beneficial to those who are working with or those who might be exposed to ultraviolet radiation. Given that light may reach the eye from the sides, full coverage eye protection is best. Ordinary, untreated glasses give some protection. Most plastic lenses give more protection than glass lenses. Some plastic lens materials, such as polycarbonate, block most UV. Most contact lenses help to protect the retina by absorbing UV radiation.

X-rays

While X-rays are used significantly in medicine, prolonged exposure to X-rays can lead to cell damage and cancer.

For example, a mammogram is an X-ray of the human breast to detect breast cancer, but if a woman starts having regular mammograms when she is too young, her chances of getting breast cancer increases.



Gamma-rays

Due to their high energies, gamma-rays are able to cause serious damage when absorbed by living cells.

Gamma-rays are not stopped by the skin and can induce DNA alteration by interfering with the genetic material of the cell. DNA double-strand breaks are generally accepted to be the most biologically significant lesion by which ionising radiation causes cancer and hereditary disease.

A study done on Russian nuclear workers exposed to external whole-body gamma-radiation at high doses shows a link between radiation exposure and death from leukaemia, lung, liver, skeletal and other solid cancers.

Cellphones and microwave radiation



Cellphone radiation and health concerns have been raised, especially following the enormous increase in their use. This is because cellphones use electromagnetic waves in the microwave range. These concerns have induced a large body of research. Concerns about effects on health have also been raised regarding other digital wireless systems, such as data communication networks. In 2009 the World Health Organisation announced that they have found a link between brain cancer and cellphones. However, there is still no firm evidence for this and the link is tenuous at best. You can find out more at <http://www.who.int/mediacentre/factsheets/fs193/en/>^a

^a<http://www.who.int/mediacentre/factsheets/fs193/en/>

Cellphone users are recommended to minimise their exposure to the radiation, by for example:

1. Use hands-free to decrease the radiation to the head.
2. Keep the mobile phone away from the body.
3. Do not use a cellphone in a car without an external antenna.

Exercise 11 - 2

1. Indicate the penetrating ability of the different kinds of EM radiation and relate it to energy of the radiation.
2. Describe the dangers of gamma rays, X-rays and the damaging effect of ultra-violet radiation on skin.

 More practice  video solutions  or help at www.everythingscience.co.za

(1.) 004t (2.) 004u

Particle-like nature of EM radiation

 ESADM

When we talk of electromagnetic radiation as a particle, we refer to photons, which are packets of energy. The energy of the photon is related to the wavelength of electromagnetic radiation according to:

DEFINITION: Planck's constant

Planck's constant is a physical constant named after Max Planck.

$$h = 6,63 \times 10^{-34} \text{ J} \cdot \text{s}$$

The energy of a photon can be calculated using the formula:

$$E = hf$$

or

$$E = h \frac{c}{\lambda},$$

where E is the energy of the photon in joules (J), h is Planck's constant, c is the speed of light, f is the frequency in hertz (Hz) and λ is the wavelength in metres (m).

The higher the frequency of EM radiation, the higher the energy.

Example 3: Calculating the energy of a photon I**QUESTION**

Calculate the energy of a photon with a frequency of 3×10^{18} Hz

SOLUTION**Step 1 : Analyse the question**

You are asked to determine the energy of a photon given the frequency. The frequency is in standard units and we know the relationship between frequency and energy.

Step 2 : Apply the equation for the energy of a photon

$$\begin{aligned} E &= hf \\ &= 6,63 \times 10^{-34} \text{ J} \cdot \text{s} \times 3 \times 10^{18} \text{ Hz} \\ &= 2 \times 10^{-15} \text{ J} \end{aligned}$$

Step 3 : Quote the final result

The energy is 2×10^{-15} J

Example 4: Calculating the energy of a photon II

QUESTION

What is the energy of an ultraviolet photon with a wavelength of 200 nm?

SOLUTION**Step 1 : Analyse the question**

You are asked to determine the energy of a photon given the wavelength. The wavelength is in standard units and we know the relationship between frequency and energy. We also know the relationship between wavelength and frequency, the equation for wave speed. The speed of light is a constant that we know.

Step 2 : Apply principles

First we determine the frequency in terms of the wavelength.

$$c = f \cdot \lambda$$

$$f = \frac{c}{\lambda}$$

We can substitute this into the equation for the energy of a photon, $E = hf$, allowing us to deduce:

$$E = h \frac{c}{\lambda}$$

Step 3 : Do the calculation

$$\begin{aligned} E &= h \frac{c}{\lambda} \\ &= (6,63 \times 10^{-34} \text{ J} \cdot \text{s}) \frac{3 \times 10^8 \text{ m} \cdot \text{s}^{-1}}{200 \times 10^{-9} \text{ m}} \\ &= 9,939 \times 10^{-10} \text{ J} \end{aligned}$$

Step 4 : Quote the final result

The energy of the photon is $9,939 \times 10^{-10} \text{ J}$

Exercise 11 - 3

1. How is the energy of a photon related to its frequency and wavelength?
2. Calculate the energy of a photon of EM radiation with a frequency of 10^{12} Hz.
3. Determine the energy of a photon of EM radiation with a wavelength of 600 nm.

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(1.) 004v (2.) 004w (3.) 004x

Animal behaviour [IKS]



People have believed that animals can predict earthquakes and other natural disasters for centuries. As early as 373 B.C., historians recorded a massive exodus of animals, including rats, snakes and weasels, from the Greek city of Helice days before a quake struck causing massive devastation.

This topic is much debated and different behaviours are sometimes seen for different kinds of animals, for example:

- **Dogs and cats:** are believed by pet owners to howl or bite their owners before natural disasters, they cite factors like a much stronger sense of smell.
- **Sharks:** researchers in Florida have reported that sharks are observed to move to deeper water before hurricanes, possibly because of a sensitivity to changes in the air pressure preceding the hurricane.
- **Rodents:** rodents that live underground will often flee their holes and burrows before a disaster. Scientists from the California Institute of Technology have noted that there are many changes preceding earthquakes such as tilting of the Earth. Rodents are often more sensitive to such small changes and will react to these changes.

- **Elephants:** will allegedly trumpet and flee to higher ground before a tsunami arrived. This is attributed to their being more sensitive to vibrations on the Earth's surface.

Many researchers argue that animals detect certain natural signals, such as the early tremblings of an earthquake, long before humans. This means that the animals have opportunity to react before we can. However it can be said that they exhibit no special understanding, they just flee as would any person hearing a shout of fire.

Another problem cited with these seemingly clairvoyant animals is that their psychic powers often are based on behaviours that people only recall after the event. Some animal behaviours happen frequently, but are not remembered unless an earthquake, tsunami, or mud slide follows. For example, if you see a dog cross a road, you just remember you saw a dog cross the road. But if an earthquake shook your neighbourhood five minutes later, would you say the dog was fleeing?

Activity:
Animals and natural disasters

Carry out research on the behaviour of animals before natural disasters.

Pick one type of natural disaster (earthquake, flood, tsunami, etc.) and see what you can find about animals reacting to that type of disaster. Ask people you know about what they have heard to get a sense of folklore.

Then research the topic to find more information and remember to critically assess all information. Things to consider:

- What scientific research has been conducted?
- Which countries does that type of disaster usually occur in?
- Do any of the native people of that country have legends/ideas about animals reacting to the disaster?
- What do people believe leads to this behaviour? i.e. do the animals have some mystic ability or are they more sensitive to anything than we are (such as low frequency radiation)

Some suggested resources for information are:

- <http://www.unep.org/ik/>
- http://earthquake.usgs.gov/learn/topics/animal_eqs.php
- <http://biology.about.com/od/animalbehavior/a/aa123104a.htm>
- http://news.nationalgeographic.com/news/2003/11/1111_031111_earthquakeanimals_2
- *Bats sing, mice giggle* by Karen Shanor and Jagmeet Kanwal
- <http://www.sheldrake.org/homepage.html>
- <http://nationalzoo.si.edu/SCBI/AnimalCare/News/earthquake.cfm>
- <http://www.animalvoice.com/animalssixthsense.htm>

Present your findings to your class. Critically analyse all the information you collect and decide what you believe.

Chapter 11 | Summary

See the summary presentation (📺 Presentation: VPfhw at www.everythingscience.co.za)

- Electromagnetic radiation has both a wave and a particle nature.
- Electromagnetic waves travel at a speed of approximately $3 \times 10^8 \text{ m} \cdot \text{s}^{-1}$ in a vacuum.
- The Electromagnetic spectrum consists of the following types of radiation: radio waves, microwaves, infrared, visible, ultraviolet, X-rays, gamma-rays.
- Gamma-rays have the most energy and are the most penetrating, while radio waves have the lowest energy and are the least penetrating.

Physical Quantities		
Quantity	Unit name	Unit symbol
Energy (E)	joule	J
Wavelength (λ)	metre	m
Period (T)	second	s
Frequency (f)	hertz	Hz (s^{-1})
Speed of light (c)	metre per second	$\text{m} \cdot \text{s}^{-1}$

Table 11.3: Units used in **electromagnetic radiation**

Chapter 11

End of chapter exercises

1. What is the energy of a photon of EM radiation with a frequency of $3 \times 10^8 \text{ Hz}$?
2. What is the energy of a photon of light with a wavelength of 660 nm?
3. What is the energy of a photon of light with a frequency of 13 THz?
4. What is the wavelength of a photon of light with a frequency of 101.3 kHz?
5. What is the energy of a photon of light with a wavelength of 532 nm and one with a frequency of 13 GHz and which has the longer wavelength?

6. List the main types of electromagnetic radiation in order of increasing wavelength.
7. List the main uses of:
 - a. radio waves
 - b. infrared
 - c. gamma rays
 - d. X-rays
8. Explain why we need to protect ourselves from ultraviolet radiation from the Sun.
9. List some advantages and disadvantages of using X-rays.
10. What precautions should we take when using cell phones?
11. Write a short essay on a type of electromagnetic waves. You should look at uses, advantages and disadvantages of your chosen radiation.
12. Explain why some types of electromagnetic radiation are more penetrating than others.

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(7.) 0051 (8.) 0052 (9.) 0053 (10.) 0054 (11.) 0055 (12.) 0056

The particles that substances are made of

12

Atoms and compounds

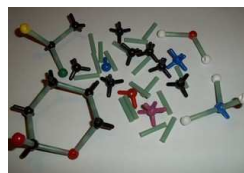
www ESADO

The atom as the building block of matter www ESADP

We have seen that different materials have different properties. But what would we find if we were to break down a material into the parts that make it up (i.e. its microscopic structure)? And how is it that this microscopic structure is able to give matter all its different properties?

See introductory video: (©) Video: VPaxx at www.everythingscience.co.za)

The answer lies in the smallest building block of matter: the **atom**. It is the *type* of atoms, and the way in which they are *arranged* in a material, that affects the properties of that substance. This is similar to building materials. We can use bricks, steel, cement, wood, straw (thatch), mud and many other things to build structures from. The choice of atoms affects the properties of matter in the same way as the choice of building material affects the properties of the structure,



It is not often that substances are found in atomic form (just as you seldom find a building or structure made from one building material). Normally, atoms are bonded (joined) to other atoms to form **compounds** or **molecules**. It is only in the *noble gases* (e.g. helium, neon and argon) that atoms are found individually and are not bonded to other atoms. We looked at some of the reasons for this in earlier chapters.

Compounds



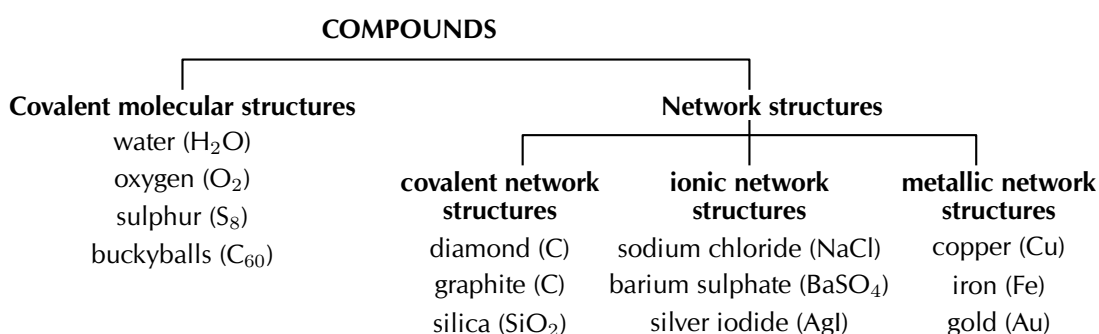
DEFINITION: Compound

A compound is a group of two or more different atoms that are attracted to each other by relatively strong forces or bonds. The atoms are combined in definite proportions.

Compounds can be divided into **molecular compounds** (molecules), **ionic compounds** (salts) and **metallic compounds** (metals).

- **Molecular compounds** form as a result of covalent bonding where electrons are shared between non-metal atoms.
- **Ionic compounds** form as a result of ionic bonding where electrons are transferred from metals to non-metals.
- **Metals** are formed as a result of metallic bonding where metal atoms lose their outer electrons to form a lattice of regularly spaced positive ions and a “pool” of delocalised electrons that surround the positive ions.

The following diagram illustrates how compounds can be subdivided by the type of bonding and the structure:



Covalent molecular structures

Relatively small molecules are called **covalent molecular structures**. These exist and interact as separate molecules. Oxygen (O_2), water (H_2O), octane (C_8H_{18}), sulphur (S_8) and buckminsterfullerene (C_{60} , buckyballs) are all examples of covalent molecular structures.

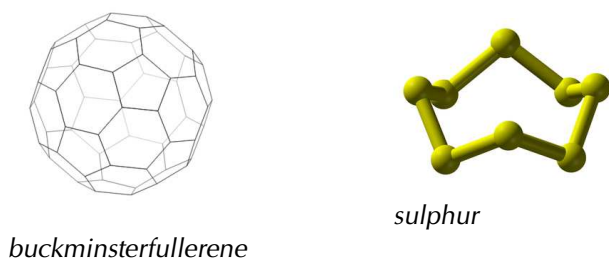


Figure 12.2: Examples of covalent molecular structures

Network structures

Compounds that exist as giant repeating lattice structures are called network structures. Examples include **covalent molecules** such as diamond, graphite and silica. **Ionic substances** are also network structures, for example a sodium chloride crystal is a huge lattice of repeating units made of sodium and chloride ions. All substances formed as a result of ionic bonding are network structures. **Metals** exist as large continuous lattice structures and are also classified as network structures. For example copper, zinc and iron can be seen as a giant crystals and are therefore considered to be network structures. ▶ See video: VPazc at www.everythingscience.co.za

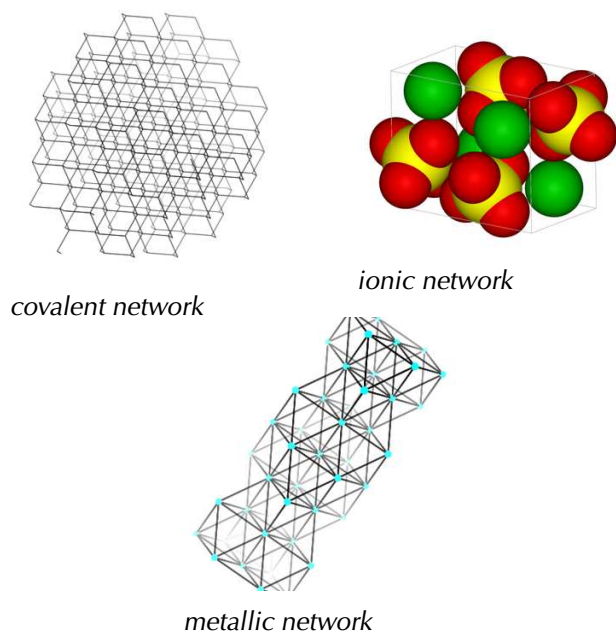


Figure 12.3: Examples of network structures

Representing molecules



The structure of a molecule can be shown in many different ways. Sometimes it is easiest to show what a molecule looks like by using different types of **diagrams**, but at other times, we may decide to simply represent a molecule using its **chemical formula** or its written name.

Using formulae to show the structure of a molecule

A **chemical formula** is an abbreviated (shortened) way of describing a compound. In chapter 2, we saw how chemical compounds can be represented using element symbols from the periodic table. A chemical formula can also tell us the *number* of atoms of each element that are in a compound and their *ratio* in that compound. If the compound is a covalent molecular compound then we can use the molecular formula.

DEFINITION: Molecular formula

The molecular formula is a concise way of expressing information about the atoms that make up a particular covalent molecular compound. The molecular formula gives the exact number of each type of atom in the molecule.

Note

The empirical and the molecular formulae can be the same. For example in carbon dioxide the molecular formula is CO_2 . This is also the empirical formula since it is the simplest ratio.

For example in figure 12.4 the molecular formula of 2-methyl propane is C_4H_{10} . This tells us that there are 4 carbon atoms and 10 hydrogen atoms in this molecule, i.e the ratio of carbon to hydrogen is 4 : 10. But we can simplify this ratio to: 2 : 5. This gives us the **empirical formula** of the molecule.

DEFINITION: Empirical formula

The empirical formula is a way of expressing the *relative* number of each type of atom in a chemical compound. The empirical formula does not show the exact number of atoms, but rather the simplest *ratio* of the atoms in the compound.

The empirical formula is useful when we want to write the formula for *network structures*. Since network structures may consist of millions of atoms, it is impossible to say exactly how many atoms are in each unit. It makes sense then to represent these units using their empirical formula. So, in the case of a metal such as copper, we would simply write Cu, or if we were to represent a unit of sodium chloride, we would simply write NaCl. Chemical formulae (i.e. the molecular or the empirical formula) therefore tell us something about the *types* of atoms that are in a compound and the *ratio* in which these atoms occur in the compound, but they don't give us any idea of what the compound actually looks like, in other words its *shape*. To show the shape of compounds we have to use diagrams. The simplest type of diagram that can be used to describe a compound is its **structural formula**. The structural formula for 2-methyl propane is shown in figure 12.4.

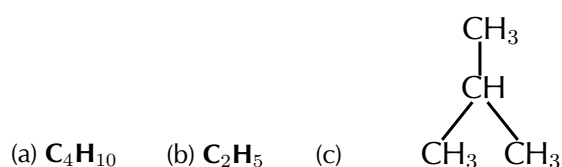


Figure 12.4: Diagram showing (a) the chemical, (b) the empirical and (c) the structural formula of 2-methyl propane

Using diagrams to show the structure of a compound

Diagrams of compounds are very useful because they help us to picture how the atoms are arranged in the compound and they help us to see the shape of the compound. There are three types of diagrams that are commonly used:

- **Wireframe or stick models**

In this model, the bonds between atoms are shown as “sticks”. These “sticks” are coloured to show which atoms are bonding.

- **Ball and stick models**

This is a 3-dimensional molecular model that uses “balls” to represent atoms and “sticks” to represent the bonds between them. The centres of the atoms (the balls) are connected by straight lines which represent the bonds between them.

- **Space-filling models**

This is also a 3-dimensional molecular model. The atoms are represented by spheres.

Table 12.1 shows examples of the different types of models for all the types of compounds.

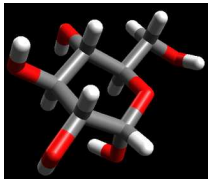
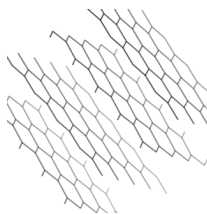
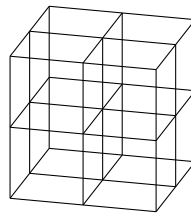
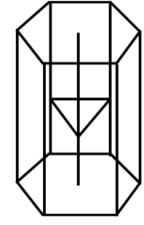
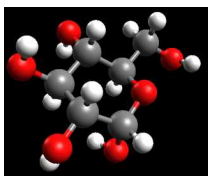
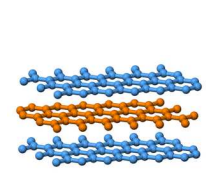
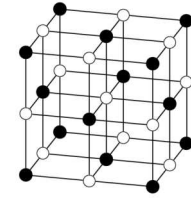
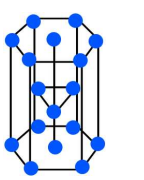
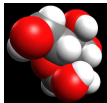
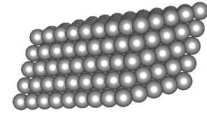
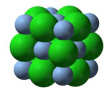
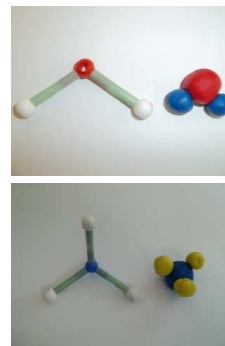
	Covalent molecular	Covalent network	Ionic network	Metallic network
Name of compound	glucose	graphite	silver chloride	zinc
Formula	$C_6H_{12}O_6$ or CH_2O	C	AgCl	Zn
Stick model				
Ball-and-stick model				
Space-filling model				

Table 12.1: Different representations for compounds

Activity:*Representing compounds*

A list of substances is given below. Make use of atomic model kits, play dough and toothpicks, or coloured polystyrene balls and skewer sticks to represent each of the substances in three dimensional structures.

- glucose ($C_6H_{12}O_6$)
- silica (SiO_2)
- sodium chloride (NaCl)
- sulphur (S_8)
- diamond (C)
- graphite (C)
- buckyballs(C_{60})
- sucrose ($C_{12}H_{22}O_{11}$)
- copper (Cu)

**Note**

CanvasMol (VPdve) is a website that allows you to view several compounds. You do not need to know these compounds, this is simply to allow you to see one way of representing compounds.

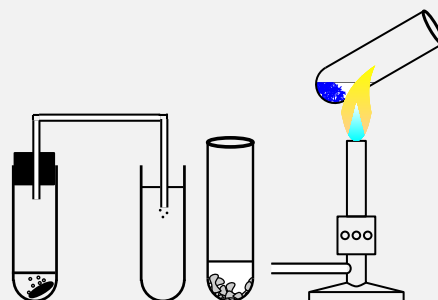
▶ See video: VPcbj at www.everythingscience.co.za

General experiment: Investigating elements and compounds

Aim: To investigate three reactions to learn about elements and compounds.

Apparatus:

- Cal-C-Vita tablet
- test tubes
- Bunsen burner
- rubber stopper
- delivery tube
- lime water (a saturated solution of Ca(OH)_2)
- candle
- matches
- copper sulphate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$)
- zinc metal
- hydrochloric acid (HCl)



Method:

Reaction 1

1. Pour clear lime water into a test tube.
2. Into a second test tube place a Cal-C-Vita tablet. Cover the tablet with water and immediately place a stopper and delivery tube into the test tube.
3. Place the other end of the delivery tube into the lime water in the first test tube. Allow it to bubble for 1-2 minutes.
4. Now remove the stopper from the second test tube and hold a lit candle at the mouth of the test tube.
5. Record your observations.

Reaction 2

1. Place a few drops of zinc metal in a test tube and cover the pieces with dilute hydrochloric acid.
2. Record your observations.
3. Now hold a burning match in the mouth of the test tube and observe what happens.

Reaction 3

1. Place a spatula full of copper sulphate crystals into a test tube and heat the tube over a Bunsen burner.
2. Record your observations.

Discussion and conclusion: In the first reaction the lime water goes milky due to the presence of carbon dioxide. (Lime water can be used to detect carbon dioxide gas.) The carbon dioxide gas comes from the sodium bicarbonate (NaHCO_3) in the tablet. When you hold the candle over the test tube, the carbon dioxide snuffs out the candle flame.

In the second reaction bubbles of hydrogen gas form. Zinc reacts with the hydrochloric acid to form zinc chloride and hydrogen gas.

In the third reaction the copper sulphate crystals go white and droplets of water form on the sides of the test tube. The copper sulphate crystals have lost their water of crystallisation.

▶ See video: VPbec at www.everythingscience.co.za

General experiment: The electrolysis of water

Aim: To investigate the elements that make up water.

Apparatus:

- water
- two pencils sharpened at both ends
- 9 volt battery
- connecting wire
- tape
- table salt or sodium sulphate

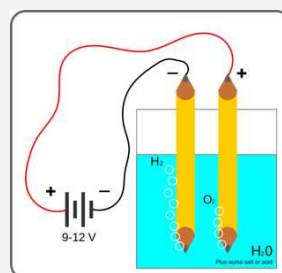


image by Nevit Dilmen

Method: Set up the apparatus as shown above. Observe what happens.

Results: You should observe bubbles forming at the tips of the pencils. Oxygen gas is formed at the positive side and hydrogen at the negative side.

Chapter 12 | Summary

See the summary presentation (📺 Presentation: VPdvy at www.everythingscience.co.za)

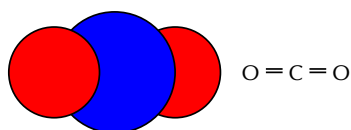
- The smallest unit of matter is the **atom**. Atoms can combine to form **compounds**.
- A **compound** is a group of two or more different atoms that are attracted to each other by relatively strong forces or bonds. The atoms are combined in definite proportions.
- In a compound, atoms are held together by **chemical bonds**. Covalent bonds, ionic bonds and metallic bonds are examples of chemical bonds.
- A **covalent bond** exists between non-metal atoms. An **ionic bond** exists between non-metal and metal atoms and a **metallic bond** exists between metal atoms.
- **Covalent molecular structures** interact and exist as separate molecules.
- **Network structures** exist as giant repeating lattices. Network structures can consist of covalent, ionic or metallic compounds.
- A chemical formula is an abbreviated (shortened) way of describing a compound.
- The **molecular formula** is a concise way of expressing information about the atoms that make up a particular covalent molecular compound. The molecular formula gives the exact number of each type of atom in the molecule.
- The **empirical formula** is a way of expressing the relative number of each type of atom in a chemical compound. The empirical formula does not show the exact number of atoms, but rather the simplest ratio of the atoms in the compound.
- The structure of a compound can be represented by stick, ball-and-stick or space-filling models.
- A **stick** model use coloured sticks to represent compounds.
- A **ball-and-stick** model is a 3-dimensional molecular model that uses “balls” to represent atoms and “sticks” to represent the bonds between them.
- A **space-filling model** is also a 3-dimensional molecular model. The atoms are represented by spheres.

Chapter 12

End of chapter exercises

1. Give one word or term for each of the following descriptions.
 - a. A composition of two or more atoms that act as a unit.
 - b. Chemical formula that gives the relative number of atoms of each element that are in a molecule.
2. Give a definition for each of the following terms:
 - a. molecule
 - b. ionic compound
 - c. covalent network structure

- d. empirical formula
 - e. ball-and-stick model
3. Ammonia, an ingredient in household cleaners, is made up of one part nitrogen (N) and three parts hydrogen (H). Answer the following questions:
- a. is ammonia a covalent, ionic or metallic substance?
 - b. write down the molecular formula for ammonia
 - c. draw a ball-and-stick diagram
 - d. draw a space-filling diagram
4. In each of the following, say whether the chemical substance is made up of covalent, molecular structures, covalent network structures, ionic network structures or metallic structures:
- a. ammonia gas (NH_3)
 - b. zinc metal (Zn)
 - c. graphite (C)
 - d. nitric acid (HNO_3)
 - e. potassium bromide (KBr)
5. Refer to the diagram below and then answer the questions that follow:



- a. Identify the molecule.
 - b. Write the molecular formula for the molecule.
 - c. Is the molecule a covalent, ionic or metallic substance? Explain.
6. Represent each of the following molecules using its *chemical formula*, *structural formula* and the *ball-and-stick model*.
- a. nitrogen
 - b. carbon dioxide
 - c. methane
 - d. argon

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Physical and chemical change

13

Introduction



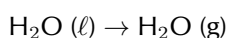
Matter is all around us. The desks we sit at, the air we breathe and the water we drink are all examples of matter. But matter doesn't always stay the same. It can change in many different ways. In this chapter, we are going to take a closer look at **physical** and **chemical** changes that occur in matter.

See introductory video: (📺 Video: VPber at www.everythingscience.co.za)

Physical changes in matter



A **physical change** is one where the particles of the substances that are involved in the change are not broken up in any way. When water is heated for example, the temperature and energy of the water molecules increases and the liquid water evaporates to form water vapour. When this happens, some kind of change has taken place, but the molecular structure of the water has not changed. This is an example of a *physical change*. All changes in state are physical changes.



Conduction (the transfer of energy through a material) is another example of a physical change. As energy is transferred from one material to another, the *energy* of each material is changed, but not its chemical makeup. Dissolving one substance in another is also a physical change.

DEFINITION: Physical change

A change that can be seen or felt, but that doesn't involve the break up of the particles in the reaction. During a physical change, the *form* of matter may change, but not its *identity*.

There are some important things to remember about physical changes in matter:

1. *Arrangement of particles* When a physical change occurs, the compounds may rearrange themselves, but the bonds in between the atoms will not break. For example when liquid water boils, the molecules will move apart but the molecule will stay intact. In other words water will not break up into hydrogen and oxygen atoms.

Figure 13.1 shows this phase change. Note that the water molecules themselves stay the same, but their arrangement changed.

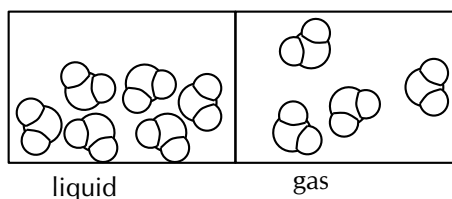


Figure 13.1: The arrangement of water molecules in the liquid and gas phase

2. *Conservation of mass*

In a physical change, the total mass, the number of atoms and the number of molecules will always stay the same. In other words you will always have the same number of molecules or atoms at the end of the change as you had at the beginning.

3. *Energy changes*

Energy changes may take place when there is a physical change in matter, but these energy changes are normally smaller than the energy changes that take place during a chemical change.

4. *Reversibility*

Physical changes in matter are usually easier to reverse than chemical changes. Methods such as filtration and distillation can be used to reverse the change. Changing the temperature is another way to reverse a physical change. For example, a mixture of salt dissolved in water can be separated by filtration, ice can be changed to liquid water and back again by changing the temperature.

FACT

The bonding of hydrogen and oxygen to form water is explosive and if the water molecule broke apart every time water boiled, life on Earth would not exist for very long!

Activity:

Physical change

Use plastic pellets or marbles to represent water in the solid state. What do you need to do to the pellets to represent the change from solid to liquid?

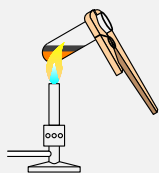
Make a mixture of sand and water. Filter this mixture. What do you observe?

Make a mixture of iron filings and sulphur. Can you separate the mixture with a magnet?

General experiment: *The synthesis of iron sulphide*

Aim: To demonstrate the synthesis of iron sulphide from iron and sulphur.

Apparatus: 5,6 g iron filings and 3,2 g powdered sulphur; porcelain dish; test tube; Bunsen burner

**Method:**

1. Measure the quantity of iron and sulphur that you need and mix them in a porcelain dish.
2. Take some of this mixture and place it in the test tube. The test tube should be about one third full.
3. Heat the test tube containing the mixture over the Bunsen burner. Increase the heat if no reaction takes place. Once the reaction begins, you will need to remove the test tube from the flame. Record your observations.
4. Wait for the product to cool before breaking the test tube with a hammer. Make sure that the test tube is rolled in paper before you do this, otherwise the glass will shatter everywhere and you may be hurt.
5. What does the product look like? Does it look anything like the original reactants? Does it have any of the properties of the reactants (e.g. the magnetism of iron)?

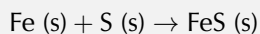
Warning:

When working with a Bunsen burner, work in a well ventilated space and ensure that there are no flammable substances close by. Always tuck loose clothing in and ensure that long hair is tied back.

Results: After you removed the test tube from the flame, the mixture glowed a bright red colour. The reaction is exothermic and *produces heats*. The product, iron sulphide, is a dark colour and does not share any of the properties of the original reactants. It is an entirely new product.

Conclusions: A synthesis reaction has taken place. The equation for the reaction

is:



Chemical Changes in Matter



When a **chemical change** takes place, new substances are formed in a chemical reaction. These new products may have very different properties from the substances that were there at the start of the reaction.

▶ See video: VPbfo at www.everythingscience.co.za

DEFINITION: Chemical change

The formation of new substances in a chemical reaction. One type of matter is changed into something different.

We will consider two examples of chemical change: the decomposition (breaking down) of hydrogen peroxide and the synthesis (forming) of water.

Decomposition of hydrogen peroxide

The decomposition (breakdown) of hydrogen peroxide (H_2O_2) to form water (H_2O) and oxygen gas (O_2) is an example of chemical change. A simplified diagram of this reaction is shown in Figure 13.2. The chemical bonds between O and H in H_2O_2 are broken and new bonds between H and O (to form H_2O) and between O and O (to form O_2) are formed. A chemical change has taken place.

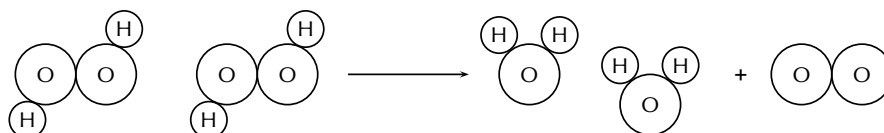
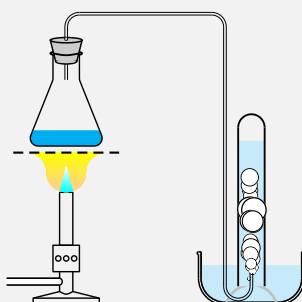


Figure 13.2: The decomposition of H_2O_2 to form H_2O and O_2

General experiment: *The decomposition of hydrogen peroxide*

Aim: To observe the decomposition of hydrogen peroxide when it is heated.

Apparatus: Dilute hydrogen peroxide (about 3%); manganese dioxide; test tubes; a water bowl; stopper and delivery tube, Bunsen burner

**Warning:**

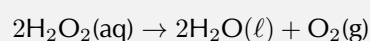
Hydrogen peroxide can cause chemical burns. Work carefully with it.

Method:

1. Put a small amount (about 5 ml) of hydrogen peroxide in a test tube.
2. Set up the apparatus as shown above.
3. Very carefully add a small amount (about 0,5 g) of manganese dioxide to the test tube containing hydrogen peroxide.

Results: You should observe a gas bubbling up into the second test tube. This reaction happens quite rapidly.

Conclusions: When hydrogen peroxide is added to manganese dioxide it decomposes to form oxygen and water. The chemical decomposition reaction that takes place can be written as follows:



Note that the manganese dioxide is a catalyst and does not take part in the reac-

Note

This experiment used the downward displacement of water to collect a gas. This is a very common way to collect a gas in chemistry. The oxygen that is evolved in this reaction moves along the delivery tube and then collects in the top of the test tube. It does this because it is less dense than water and does not dissolve in water, so the water is displaced downwards.

FACT

This reaction is often performed without collecting the oxygen gas and is commonly known as the elephant's toothpaste reaction.

tion, so we do not show it in the balanced equation. (A catalyst helps speed up a chemical reaction.)

▶ See video: VPbfq at www.everythingscience.co.za

The above experiment can be very vigorous and produce a lot of oxygen very rapidly. For this reason you should use dilute hydrogen peroxide and only a small amount of manganese dioxide.

The synthesis of water

The synthesis (forming) of water (H_2O) from hydrogen gas (H_2) and oxygen gas (O_2) is another example of chemical change. A simplified diagram of this reaction is shown in figure 13.3. The chemical bonds between O in O_2 and between H in H_2 are broken and new bonds between H and O (to form H_2O) are formed. A chemical change has taken place.

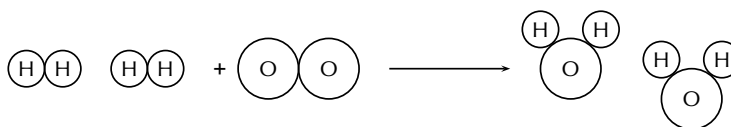


Figure 13.3: The synthesis of H_2O from H_2 and O_2

▶ See video: VPbhy at www.everythingscience.co.za

General experiment: The synthesis of water**Aim:**

To observe the synthesis of water.

Apparatus:

Hydrogen gas; balloon; string; candle; long stick; ear plugs; safety glasses



photos by argonnenationallaboratory on Flickr.com

Method:

1. Half fill a balloon with hydrogen gas.
2. Fill the remainder of the balloon with oxygen gas. (You can also just use your breath to fill the balloon.)
3. Tie the balloon to one end of the string. Tie down the other end of the string so that the balloon is positioned in mid air, away from any people, objects, walls, ceilings etc.
4. Attach the candle tightly to the stick and light the candle.
5. Standing away from the balloon, carefully hold the candle to the balloon.

Warning:

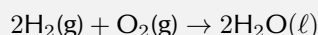
This reaction can be highly explosive, for this reason it is best done outdoors. Always ensure that you wear ear protection or block your ears.
Always have more oxygen than hydrogen in the balloon.

Results:

When you bring the candle close to the balloon you should see a flame and hear a loud bang.

Conclusions:

When a mixture of hydrogen and oxygen gas is set alight with a candle a chemical change occurs. Water is made according to the following equation:

**FACT**

A mixture of hydrogen and oxygen gas is used as a fuel to get rockets into space.

There are some important things to remember about chemical changes:

1. Arrangement of particles

During a chemical change, the particles themselves are changed in some way. In the example of hydrogen peroxide that was used earlier, the H_2O_2 molecules were split up into their component atoms. The number of particles will change because each H_2O_2 molecule breaks down into two water molecules (H_2O) and one oxygen molecule (O_2).

2. Energy changes

The energy changes that take place during a chemical reaction are much greater than those that take place during a physical change in matter. During a chemical reaction, energy is used up in order to break bonds and then energy is released when the new

product is formed.

3. Reversibility

Chemical changes are far more difficult to reverse than physical changes. When hydrogen peroxide decomposes into water and oxygen, it is almost impossible to get back to hydrogen peroxide.

4. Mass conservation

Mass is conserved during a chemical change, but the number of molecules may change. In the example of the decomposition of hydrogen peroxide, for every two molecules of hydrogen peroxide that decomposes, three molecules are formed (two water and one oxygen).

Table 13.1 highlights these concepts for the decomposition of hydrogen peroxide.

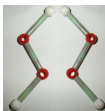
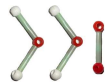
	$2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$  	
Molecules	two molecules	three molecules
Energy changes	energy taken in when bonds are broken	energy given off when bonds are formed
Mass is conserved	$4(1,01) + 4(16,0) = 68,04$	$2(18,02) + 2(16,0) = 68,04$
Atoms are conserved	4 oxygen atoms, 4 hydrogen atoms	4 oxygen atoms, 4 hydrogen atoms

Table 13.1: Important concepts in chemical change

Exercise 13 - 1

For each of the following say whether a chemical or a physical change occurs.

1. Melting candle wax.
2. Mixing sodium chloride (NaCl) and silver nitrate (AgNO₃) to form silver chloride (AgCl).
3. Mixing hydrochloric acid (HCl) and magnesium ribbon (Mg) to form magnesium chloride (MgCl₂).
4. Dissolving salt in water.
5. Tearing a piece of magnesium ribbon.

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Conservation of atoms and mass in reactions



In a chemical reaction the total mass of all the substances taking part in the reaction remains the same. Also, the number of **atoms** in a reaction remains the same. Mass cannot be created or destroyed in a chemical reaction.

▶ See video: VPbkj at www.everythingscience.co.za

DEFINITION: Law of conservation of mass

The law of conservation of mass states that the total mass of substances taking part in a chemical reaction is conserved during the reaction.

Table 13.1 illustrates this law for the decomposition of hydrogen peroxide.

Activity:

The conservation of atoms in chemical reactions

We will use the reaction of hydrogen and oxygen to form water in this activity.

Materials: Coloured modelling clay rolled into balls or marbles and prestik to represent atoms. Each colour will represent a different element.

Method:

1. Build your reactants. Use marbles and prestik or modelling clay to represent the reactants and put these on one side of your table. Make at least ten (H_2) units and at least five (O_2) units.
2. Place the H_2 and O_2 units on a table. The table represents the “test tube” where the reaction is going to take place.
3. Now count the number of atoms (H and O) you have in your “test tube”. Fill in the reactants column in the table below. Refer to table 13.1 to help you fill in the mass row.
4. Let the reaction take place. Each person can now take the H and O unit and use them to make water units. Break the H and O units apart and build H_2O units with the parts. These are the products. Place the products on the table.

5. When the “reaction” has finished (i.e. when all the H and O units have been used) count the number of atoms (H and O) and complete the table.
6. What do you notice about the number of atoms for the reactants, compared to the products?
7. Write a balanced equation for this reaction and use your models to build this equation.

	Reactants	Products
		
Number of molecules		
Mass		
Number of atoms		

Discussion: You should have noticed that the number of atoms in the reactants is the same as the number of atoms in the product. The number of atoms is conserved during the reaction. However, you will also see that the number of molecules in the reactants and products are not the same. The number of molecules is not conserved during the reaction.

Informal experiment: Conservation of matter

Aim: To prove the law of conservation of matter experimentally.

Materials:

Reaction 1:

3 beakers; silver nitrate; sodium iodide; mass meter

Reaction 2:

hydrochloric acid; bromothymol blue; sodium hydroxide solution; mass meter

Reaction 3:

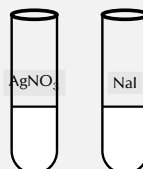
any effervescent tablet (e.g. Cal-C-Vita tablet), balloon; rubber band; mass meter; test tube; beaker

Warning:

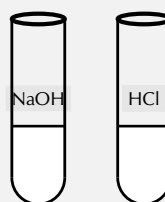
Always be careful when handling chemicals (particularly strong acids like hydrochloric acid) as you can burn yourself badly.

Method:*Reaction 1*

1. Solution 1: In one of the beakers dissolve 5 g of silver nitrate in 100 ml of water.
2. Solution 2: In a second beaker, dissolve 4.5 g of sodium iodide in 100 ml of water.
3. Determine the mass of each of the reactants.
4. Add solution 1 to solution 2. What do you observe?
Has a chemical reaction taken place?
5. Determine the mass of the products.
6. What do you notice about the masses?
7. Write a balanced equation for this reaction.

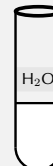
*Reaction 2:*

1. Solution 1: Dissolve 0.4 g of sodium hydroxide in 100 ml of water. Add a few drops of bromothymol blue indicator to the solution.
2. Solution 2: Measure 100 ml of 0,1 M hydrochloric acid solution into a second beaker.
3. Determine the mass of the reactants.
4. Add small quantities of solution 2 to solution 1 (you can use a plastic pipette for this) until a colour change has taken place. Has a chemical reaction taken place?
5. Determine the mass of hydrochloric acid added. (You do this by weighing the remaining solution and subtracting this from the starting mass)
6. Compare the mass before the reaction to the total mass after the reaction. What do you notice?
7. Write a balanced equation for this reaction.



Reaction 3

1. Half fill a large test tube with water.
2. Determine the mass of the test tube and water.
3. Break an effervescent tablet in two or three pieces and place them in a balloon.
4. Determine the mass of the balloon and tablet.
5. Fit the balloon tightly to the test tube, being careful to not drop the contents into the water. You can stand the test tube in a beaker to help you do this.
6. Determine the total mass of the test tube and balloon.
7. Lift the balloon so that the tablet goes into the water. What do you observe? Has a chemical reaction taken place?
8. Determine the mass of the test tube balloon combination.
9. What do you observe about the masses before and after the reaction?



Results: Fill in the following table for the total mass of reactants (starting materials) and products (ending materials).

	Reaction 1	Reaction 2	Reaction 3
Reactants			
Products			

Add the masses for the reactants for each reaction. Do the same for the products. For each reaction compare the mass of the reactants to the mass of the products. What do you notice? Is the mass conserved?

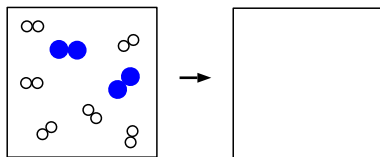
In the experiment above you should have found that the total mass at the start of the reaction is the same as the mass at the end of the reaction. Mass does not appear or disappear in chemical reactions. Mass is conserved, in other words, the total mass you start with is the total mass you will end with.

Exercise 13 - 2

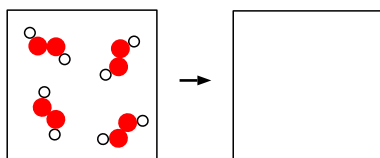
Complete the following chemical reactions to show that atoms and mass are

conserved. For each reaction give the total molecular mass of the reactants and the products.

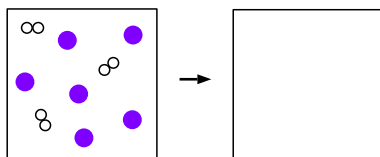
1. Hydrogen gas combines with nitrogen gas to form ammonia.



2. Hydrogen peroxide decomposes (breaks down) to form hydrogen and oxygen.



3. Calcium and oxygen gas react to form calcium oxide.



More practice video solutions or help at www.everythingscience.co.za

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Law of constant composition

ESADW

In any given chemical compound, the elements always combine in the same proportion with each other. This is the **law of constant composition**.

The **law of constant composition** says that, in any particular chemical compound, all samples of that compound will be made up of the same elements in the same proportion or ratio. For example, any water molecule is always made up of two hydrogen atoms and one oxygen atom in a 2 : 1 ratio. If we look at the relative masses of oxygen and hydrogen in

a water molecule, we see that 94% of the mass of a water molecule is accounted for by oxygen and the remaining 6% is the mass of hydrogen. This mass proportion will be the same for any water molecule.

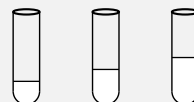
This does not mean that hydrogen and oxygen always combine in a 2 : 1 ratio to form H_2O . Multiple proportions are possible. For example, hydrogen and oxygen may combine in different proportions to form H_2O_2 rather than H_2O . In H_2O_2 , the H : O ratio is 1 : 1 and the mass ratio of hydrogen to oxygen is 1 : 16. This will be the same for any molecule of hydrogen peroxide.

Investigation: Law of constant composition

Aim: To investigate the ratio in which compounds combine.

Apparatus:

- 0,1 M silver nitrate (AgNO_3)
- 0,1 M sodium chloride (NaCl)
- 0,1 M lead nitrate (PbNO_3)
- 0,1 M sodium iodide (NaI)
- 0,1 M iron (III) chloride (FeCl_3)
- 0,1 M sodium hydroxide (NaOH)
- 9 large test tubes
- 3 propettes



Method:

Reaction 1: Prepare three test tubes with 5 ml, 10 ml and 15 ml of silver nitrate respectively. Using a clean propette add 5 ml of sodium chloride to each one and observe what happens.

Reaction 2: Prepare three test tubes with 5 ml, 10 ml and 15 ml of lead nitrate respectively. Using a clean propette add 5 ml of sodium iodide to each one and observe what happens. Write a balanced equation for this reaction.

Reaction 3: Prepare three test tubes with 5 ml, 10 ml and 15 ml of sodium hydroxide respectively. Add 5 ml of iron(III) chloride to each one and observe what happens.

Discussion and conclusion: Regardless of the amount of reactants added, the same products, with the same compositions, are formed (i.e. the precipitate observed in the reactions). However, if the reactants are not added in the correct ratios, there will be unreacted reactants that

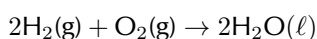
will remain in the final solution, together with the products formed.

Volume relationships in gases

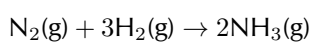


In a chemical reaction between gases, the relative volumes of the gases in the reaction are present in a ratio of small whole numbers if all the gases are at the same temperature and pressure. This relationship is also known as **Gay-Lussac's Law**.

For example, in the reaction between hydrogen and oxygen to produce water, two volumes of H_2 react with 1 volume of O_2 to produce 2 volumes of H_2O .



In the reaction to produce ammonia, one volume of nitrogen gas reacts with three volumes of hydrogen gas to produce two volumes of ammonia gas.



Chapter 13 | Summary

See the summary presentation (📺 Presentation: VPdwh at www.everythingscience.co.za)

- Matter does not stay the same. It may undergo physical or chemical changes.
- A **physical change** is a change that can be seen or felt, but that does not involve the break up of the particles in the reaction. During a physical change, the form of matter may change, but not its identity.
- During a physical change, the **arrangement of particles** may change but the mass, number of atoms and number of molecules will stay the same.
- Physical changes involve small changes in **energy** and are easily reversible.
- A chemical change occurs when one or more substances change into other materials. A chemical reaction involves the formation of new substances with **different properties**. For example, hydrogen and oxygen react to form water
- A chemical change may involve a **decomposition** or **synthesis** reaction. During a chemical change, the mass and number of atoms is conserved, but the number of

molecules is not always the same.

- Chemical reactions involve large changes in energy. Chemical reactions are not easily reversible.
- The **law of conservation of mass** states that the total mass of all the substances taking part in a chemical reaction is conserved and the number of atoms of each element in the reaction does not change when a new product is formed.
- The **law of constant composition** states that in any particular compound, all samples of that compound will be made up of the same elements in the same proportion or ratio.
- **Gay-Lussac's Law** states that in a chemical reaction between gases, the relative volumes of the gases in the reaction are present in a ratio of small whole numbers if all the gases are at the same temperature and pressure.

Chapter 13

End of chapter exercises

1. For each of the following definitions give one word or term:
 - a. A change that can be seen or felt, where the particles involved are not broken up in any way
 - b. The formation of new substances in a chemical reaction
 - c. A reaction where a new product is formed from elements or smaller compounds
2. Explain how a chemical change differs from a physical change.
3. Complete the following table by saying whether each of the descriptions is an example of a physical or chemical change:

Description	Physical or chemical
hot and cold water mix together	
milk turns sour	
a car starts to rust	
food digests in the stomach	
alcohol disappears when it is placed on your skin	
warming food in a microwave	
separating sand and gravel	
fireworks exploding	

4. For each of the following reactions, say whether it is an example of a synthesis or decomposition reaction:

- a. $(\text{NH}_4)_2\text{CO}_3 \rightarrow \text{NH}_3 + \text{CO}_2 + \text{H}_2\text{O}$
b. $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$
c. $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
5. For the following equation: $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO} + \text{CO}_2$ show that the law of conservation of mass applies. Draw sub-microscopic diagrams to represent this reaction.

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(1.) 005b (2.) 005c (3.) 005d (4.) 005e (5.) 005f

Representing chemical change

14

Introduction



As we have already mentioned, a number of changes can occur when elements are combined with one another. These changes may either be *physical* or *chemical*. In this chapter we will look at chemical changes. One way of representing chemical changes is through **balanced chemical equations**. A chemical equation describes a chemical reaction by using symbols for the elements involved. For example, if we look at the reaction between iron (Fe) and sulphur (S) to form iron sulphide (FeS), we could represent these changes in a sentence, in a word equation or using chemical symbols:

Sentence: Iron reacts with sulphur to form iron sulphide. **Word equation:** Iron + sulphur → iron sulphide. **Chemical symbols:** $\text{Fe} + \text{S} \rightarrow \text{FeS}$

Another example would be:

Sentence: Ammonia reacts with oxygen to form nitrogen monoxide and water. **Word equation:** Ammonia + oxygen → nitrogen monoxide + water. **Chemical symbols:** $4\text{NH}_3 + 5\text{O}_2 \rightarrow 4\text{NO} + 6\text{H}_2\text{O}$

See introductory video: (▶ Video: VPbkm at www.everythingscience.co.za)

Compounds on the left of the arrow are called the **reactants** and these are needed for the reaction to take place. The compounds on the right are called the **products** and these are what is formed from the reaction.

In order to be able to write a balanced chemical equation, there are a number of important things that need to be done:

1. Know the chemical symbols for the elements involved in the reaction
2. Be able to write the chemical formulae for different reactants and products
3. Balance chemical equations by understanding the laws that govern chemical change
4. Know the state symbols for the equation

We will look at each of these steps separately in the next sections.

Chemical symbols



It is very important to know the chemical symbols for common elements in the periodic table, so that you are able to write chemical equations and to recognise different compounds.

Activity:*Revising common chemical symbols*

- Write down the chemical symbols and names of all the elements that you know.
- Compare your list with another learner and add any symbols and names that you don't have.
- Know the symbols for at least the first thirty six elements in the periodic table. You should also learn the symbols for other common elements that are not in the first thirty six.
- Set a short test on naming elements and compounds for someone else in the class and then exchange tests with them so that you each have the chance to answer a test.

Writing chemical formulae



A **chemical formula** is a concise way of giving information about the atoms that make up a particular chemical compound. A chemical formula shows each element by its symbol and also shows how many atoms of each element are found in that compound. The number of atoms (if greater than one) is shown as a subscript.

The following exercise serves as revision. If you do not recall how to write chemical formulae refer back to chapter 2.

Exercise 14 - 1

1. Write down the chemical formula for each of the following compounds:

- | | |
|--------------------------------|----------------------------------|
| a. iron (III) chloride | acts with oxygen |
| b. zinc nitrate | g. the product when hydrogen re- |
| c. aluminium sulphate | acts with nitrogen |
| d. calcium hydroxide | h. potassium oxide |
| e. magnesium carbonate | i. copper (II) bromide |
| f. the product when carbon re- | j. potassium dichromate |

2. Write down the name for each of the following compounds:

- | | |
|---------------------------------|--------------------------------|
| a. SO_2 | d. BaF_2 |
| b. KMnO_4 | e. $\text{Cr}(\text{HSO}_4)_3$ |
| c. $(\text{NH}_4)_2\text{SO}_4$ | f. CH_4 |

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(1.) 02u2 (2.) 02u3

Balancing chemical equations

 **ESAEB**

The law of conservation of mass

 **ESAEC**

In order to balance a chemical equation, it is important to understand the law of conservation of mass.

DEFINITION: *The law of conservation of mass*

The mass of a closed system of substances will remain constant, regardless of the processes acting inside the system. Matter can change form, but cannot be created or destroyed.

For any chemical equation (in a closed system) the **mass** of the reactants must be equal to the mass of the products. In order to make sure that this is the case, the number of **atoms** of each element in the reactants must be equal to the number of atoms of those same elements in the products. An example is shown below:

		
$\text{Fe} + \text{S} \rightarrow \text{FeS}$		
Mass of one atom of Fe is 55,8	Mass of one atom of S is 32,1	Mass of one atom of FeS is 87,9
Mass of reactants is 87,9		Mass of products is 87,9

Tip

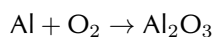
Iron is a metal. When we represent it in a balanced chemical equation, we write only Fe. Sulphur occurs as S₈ but we write only the empirical formula: S. We do this for all network structures. Writing formulae like this represents *one unit* of the compound or network structure.

To calculate the **mass of the molecules** we use the relative atomic masses for iron and sulphur, as seen in table 14.2. You will notice that the mass of the reactants equals the mass of the product. A chemical equation that is **balanced** will always reflect the **law of conservation of mass** and the **law of conservation of atoms**.

Activity:*Balancing chemical equations*

1. You will need: coloured balls (or marbles), prestik, a sheet of paper and coloured pens.

We will try to balance the following equation:



Take one ball of one colour. This represents a molecule of Al. Take two balls of another colour and stick them together. This represents a molecule of O₂. Place these molecules on your left. Now take two balls of the first colour and three balls of the second colour to form Al₂O₃. Place this compound on your right. On a piece of paper draw coloured circles to represent the balls. Draw a line down the centre of the paper to represent the molecules on the left and on the right.

Count the number of balls on the left and the number on the right. Do you

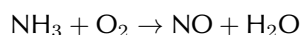
have the same number of each colour on both sides? If not, the equation is not balanced. How many balls of each colour will you have to add to each side to make the number of balls the same? How would you add these balls?

You should find that you need four balls of the first colour for Al and three pairs of balls of the second colour (i.e. six balls in total) for O₂ on the left side. On the right side you should find that you need 2 clusters of balls for Al₂O₃. We say that the balanced equation is:

$4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$ **2.** Use jelly tots and toothpicks to build the following chemical equation. Make sure that your atoms are balanced. Use the same colour jelly tots for the same atoms.



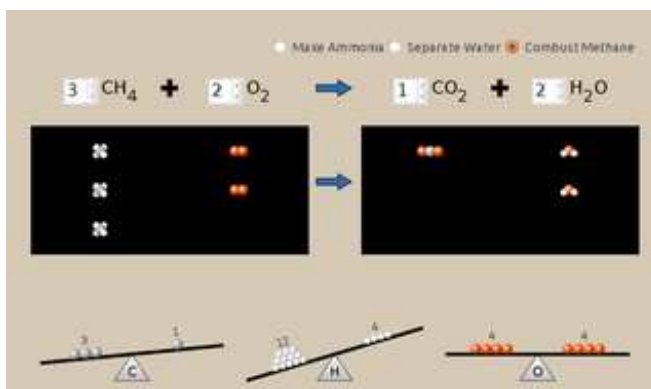
Add compounds until the atoms are balanced. Write the equation down and use a coefficient to indicate how many compounds you used. For example if you had to use three water molecules then write 3H₂O **3** Use ball and stick drawings to balance the atoms in the following reaction:



Use your drawings to write a balanced chemical equation for the reaction.

4 Lead (Pb), lead (IV) oxide (PbO₂) and sulphuric acid (H₂SO₄) are used in car batteries. The following reaction takes place: $\text{Pb} + \text{PbO}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{PbSO}_4 + \text{H}_2\text{O}$

Cut out circles from four different colours of paper to represent each of the atoms. Build a few of the compounds (Pb PbO₂ H₂SO₄). These are the reactants. Do not build the products. Rearrange the atoms so that the products are formed. Add more reactants if needed to balance the atoms (e.g. you will need two H₂SO₄ molecules). Use what you have learnt to write a balanced equation for the reaction.



Steps to balance a chemical equation through inspection

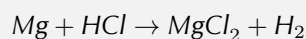
When balancing a chemical equation, there are a number of steps that need to be followed.

- Step 1:** Identify the reactants and the products in the reaction and write their chemical formulae.
- Step 2:** Write the equation by putting the reactants on the left of the arrow and the products on the right.
- Step 3:** Count the number of atoms of each element in the reactants and the number of atoms of each element in the products.
- Step 4:** If the equation is not balanced, change the coefficients of the molecules until the number of atoms of each element on either side of the equation balance.
- Step 5:** Check that the atoms are in fact balanced.
- Step 6:** (we will look at this a little later): Add any extra details to the equation e.g. phase symbols.

Example 1: Balancing chemical equations 1

QUESTION

Balance the following equation:



SOLUTION

Step 1 : Identify the reactants and products

This has been done in the question.

Step 2 : Write the equation for the reaction

This has been done in the question.

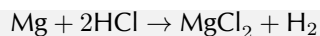
Step 3 : Count the number of atoms of each element in the reactants and products

Reactants: Mg = 1 atom; H = 1 atom; Cl = 1 atom

Products: Mg = 1 atom; H = 2 atoms; Cl = 2 atoms

Step 4 : Balance the equation

The equation is not balanced since there are two chlorine atoms in the product and only one in the reactants. If we add a coefficient of two to the HCl to increase the number of H and Cl atoms in the reactants, the equation will look like this:



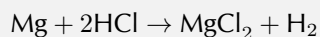
Step 5 : **Check that the atoms are balanced**

If we count the atoms on each side of the equation, we find the following:

Reactants: Mg = 1; H = 2; Cl = 2

Products: Mg = 1; H = 2; Cl = 2

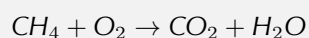
The equation is balanced. The final equation is:



Example 2: Balancing chemical equations 2

QUESTION

Balance the following equation:



SOLUTION

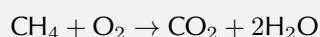
Step 1 : **Count the number of atoms of each element in the reactants and products**

Reactants: C = 1; H = 4; O = 2

Products: C = 1; H = 2; O = 3

Step 2 : **Balance the equation**

If we add a coefficient of 2 to H_2O , then the number of hydrogen atoms in the products will be 4, which is the same as for the reactants. The equation will be:

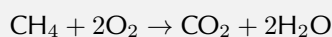


Step 3 : **Check that the atoms balance**

Reactants: C = 1; H = 4; O = 2

Products: C = 1; H = 4; O = 4

You will see that, although the number of *hydrogen* atoms now balances, there are more oxygen atoms in the products. You now need to repeat the previous step. If we put a coefficient of 2 in front of O₂, then we will increase the number of oxygen atoms in the reactants by 2. The new equation is:



When we check the number of atoms again, we find that the number of atoms of each element in the reactants is the same as the number in the products. The equation is now balanced.

Example 3: Balancing chemical equations 3

QUESTION

In our bodies, sugar (C₆H₁₂O₆) reacts with the oxygen we breathe in to produce carbon dioxide, water and energy. Write the balanced equation for this reaction.

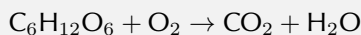
SOLUTION

Step 1 : Identify the reactants and products in the reaction.

Reactants: sugar (C₆H₁₂O₆) and oxygen (O₂)

Products: carbon dioxide (CO₂) and water (H₂O)

Step 2 : Write the equation



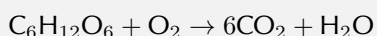
Step 3 : Count the number of atoms of each element in the reactants and in the products

Reactants: C = 6; H = 12; O = 8

Products: C = 1; H = 2; O = 3

Step 4 : Balance the equation

It is easier to start with carbon as it only appears once on each side. If we add a 6 in front of CO₂, the equation looks like this:

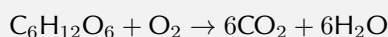


Reactants: C = 6; H = 12; O = 8

Products: C = 6; H = 2; O = 13

Step 5 : Change the coefficients again to try to balance the equation.

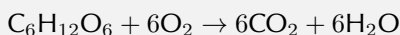
Let us try to get the number of hydrogens the same this time.



Reactants: C = 6; H = 12; O = 8

Products: C = 6; H = 12; O = 18

Step 6 : Now we just need to balance the oxygen atoms.



Reactants: C = 6; H = 12; O = 18

Products: C = 6; H = 12; O = 18

See simulation: (🎮) Simulation: VPee at www.everythingscience.co.za

Exercise 14 - 2

Balance the following equations:

1. $\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$
2. $\text{Ca} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{H}_2$
3. $\text{CuCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{CuSO}_4 + \text{H}_2\text{O} + \text{CO}_2$
4. $\text{CaCl}_2 + \text{Na}_2\text{CO}_3 \rightarrow \text{CaCO}_3 + \text{NaCl}$
5. $\text{C}_{12}\text{H}_{22}\text{O}_{11} + \text{O}_2 \rightarrow \text{H}_2\text{O} + \text{CO}_2$
6. Barium chloride reacts with sulphuric acid to produce barium sulphate and hydrochloric acid.
7. Ethane (C_2H_6) reacts with oxygen to form carbon dioxide and steam.
8. Ammonium carbonate is often used as a smelling salt. Balance the following reaction for the decomposition of ammonium carbonate: $(\text{NH}_4)_2\text{CO}_3(\text{s}) \rightarrow \text{NH}_3(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\ell)$
9. Hydrogen fuel cells are extremely important in the development of alternative energy sources. Many of these cells work by reacting hydrogen and oxygen gases together to form water, a reaction which also produces electricity. Balance the following equation: $\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\ell)$
10. The synthesis of ammonia (NH_3), made famous by the German chemist Fritz Haber in the early 20th century, is one of the most important reactions in the chemical industry. Balance the following equation used to produce ammonia: $\text{N}_2(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{NH}_3(\text{g})$

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(1.) 005g (2.) 005h (3.) 005i (4.) 005j (5.) 005k (6.) 005m
(7.) 005n (8.) 005p (9.) 005q (10.) 005r

State symbols and other information



The state (phase) of compounds can be expressed in the chemical equation. This is done by placing the correct label on the right hand side of the formula. The following four labels can be used:

1. (g) for gaseous compounds
2. (l) for liquids
3. (s) for solid compounds
4. (aq) for an aqueous (water) solution

To show that heat is needed for a reaction, a Greek delta (Δ) is placed above the arrow. For example: $\text{NH}_4\text{Cl} \xrightarrow{\Delta} \text{NH}_3 + \text{HCl}$

Example 4: Balancing chemical equations 4

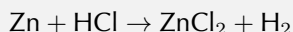
QUESTION

Solid zinc metal reacts with aqueous hydrochloric acid to form an aqueous solution of zinc chloride (ZnCl_2) and hydrogen gas. Write a balanced equation for this reaction.

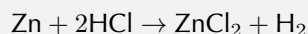
SOLUTION

Step 1 : Identify the reactants and products

The reactants are zinc (Zn) and hydrochloric acid (HCl). The products are zinc chloride (ZnCl_2) and hydrogen (H_2).

Step 2 : Write the equation**Step 3 : Balance the equation**

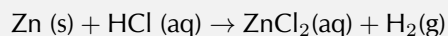
You will notice that the zinc atoms balance but the chlorine and hydrogen atoms do not. Since there are two chlorine atoms on the right and only one on the left, we will give HCl a coefficient of 2 so that there will be two chlorine atoms on each side of the equation.

**Step 4 : Check that all the atoms balance**

When you look at the equation again, you will see that all the atoms are now balanced.

Step 5 : Ensure all details (e.g. state symbols) are added

In the initial description, you were told that zinc was a metal, hydrochloric acid and zinc chloride were in aqueous solutions and hydrogen was a gas.



▶ See video: VPemn at www.everythingscience.co.za

Exercise 14 - 3

Write balanced equations for each of the following reactions, include state symbols:

1. Lead (II) nitrate solution reacts with a potassium iodide solution to form a precipitate (solid) of lead iodide while potassium nitrate remains in solution.
2. When heated, aluminium metal reacts with solid copper oxide to produce copper metal and aluminium oxide (Al_2O_3).
3. When calcium chloride solution is mixed with silver nitrate solution, a white precipitate (solid) of silver chloride appears. Calcium nitrate ($\text{Ca}(\text{NO}_3)_2$) is also produced in the solution.
4. Solid ammonium carbonate decomposes to form three gaseous products.



More practice



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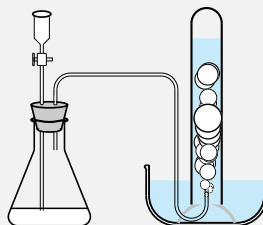
(1.) 00b7 (2.) 00b8 (3.) 00b9 (4.) 00ba

General experiment: *The relationship between product and reactant*

Aim: To investigate the relationship between the amount of product and the amount of reactant.

Apparatus:

- flask
- measuring cylinder
- water bowl
- delivery tube
- funnel with stopcock
- stopper
- sodium hydrogen carbonate (NaHCO_3) powder
- dilute sulphuric acid (H_2SO_4)



Method:

- 1 Weigh 20 g of NaHCO_3 and place it into a flask.
- 2 Set up the above apparatus.
- 3 Measure out 5 ml of H_2SO_4 and carefully pour this into the funnel (make sure that the stopcock is closed).
- 4 Slowly add the H_2SO_4 to the NaHCO_3 by opening the stopcock.
- 5 Observe what happens.
- 6 Record the volume of gas collected in the measuring cylinder.
- 7 Repeat the above steps but this time use 10 ml of H_2SO_4 .
- 8 Write a balanced equation for this reaction. (Hint: carbon dioxide gas is formed, as well as water and sodium sulphate.)

Results and discussion: You should observe that more gas is formed when using more H_2SO_4 .

Chapter 14 | Summary

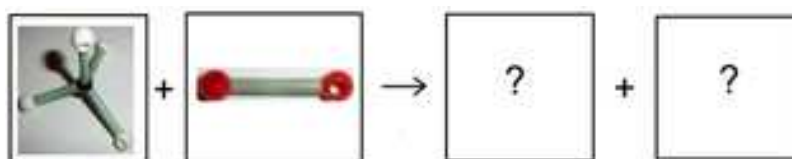
See the summary presentation (Presentation: VPeca at www.everythingscience.co.za)

- A **chemical equation** uses symbols to describe a chemical reaction.
- In a chemical equation, **reactants** are written on the left hand side of the equation and the **products** on the right. The arrow is used to show the direction of the reaction.
- When representing chemical change, it is important to be able to write the **chemical formula** of a compound.
- The law of conservation of mass states that the mass of a closed system of substances will remain constant, regardless of the processes acting inside the system. Matter can change form, but cannot be created or destroyed.
- In any chemical reaction, the **law of conservation of mass** applies. This means that the total atomic mass of the reactants must be the same as the total atomic mass of the products. This also means that the total number of atoms of the reactants must be the same as the total number of atoms of the product.
- If the number of atoms of each element in the reactants is the same as the number of atoms of each element in the product, then the equation is **balanced**.
- If the number of atoms of each element in the reactants is not the same as the number of atoms of each element in the product, then the equation is **not balanced**.
- In order to balance an equation, **coefficients** can be placed in front of the reactants and products until the number of atoms of each element is the same on both sides of the equation.
- The state of the compounds in a chemical reaction can be expressed in the chemical equation by using one of four symbols. The symbols are g (gas), *l* (liquid), s (solid) and aq (aqueous solutions). These symbols are written in brackets after the compound.

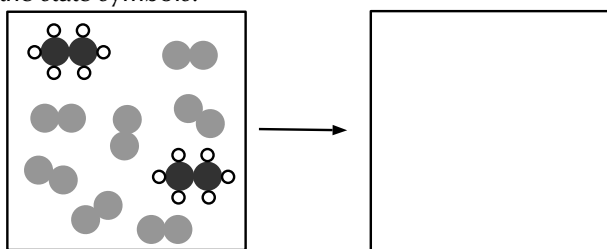
Chapter 14

End of chapter exercises

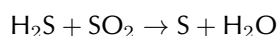
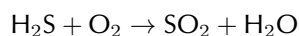
1. Propane is a fuel that is commonly used as a heat source for engines and homes. Balance the following equation for the combustion of propane:
$$\text{C}_3\text{H}_8(\ell) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\ell)$$
2. Methane (CH_4) burns in oxygen according to the following reaction.



- a. Complete the diagrams by drawing ball-and-stick models of the products.
 - b. Write a balanced chemical equation for the reaction and include state symbols.
3. Chemical weapons were banned by the Geneva Protocol in 1925. According to this protocol, all chemicals that release suffocating and poisonous gases are not to be used as weapons. White phosphorus, a very reactive allotrope of phosphorus, was recently used during a military attack. Phosphorus burns vigorously in oxygen. Many people got severe burns and some died as a result. The equation for this spontaneous reaction is: $\text{P}_4(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{P}_2\text{O}_5(\text{s})$
 - a. Balance the chemical equation.
 - b. Prove that the law of conservation of mass is obeyed during this chemical reaction.
 - c. Name the product formed during this reaction.
 - d. Classify the reaction as a synthesis or decomposition reaction. Give a reason for your answer.
4. The following diagrams represent the combustion of ethane (C_2H_6). Complete the diagrams and write a balanced equation for the reaction. Indicate the state symbols.



5. Balance the following chemical equation:
 $\text{N}_2\text{O}_5 \rightarrow \text{NO}_2 + \text{O}_2$
 Draw submicroscopic diagrams to represent this reaction.
6. Sulphur can be produced by the Claus process. This two-step process involves reacting hydrogen sulphide with oxygen and then reacting the sulphur dioxide that is produced with more hydrogen sulphide. The equations for these two reactions are:



Balance these two equations.

7. Aspartame, an artificial sweetener, has the formula $\text{C}_{14}\text{H}_{18}\text{N}_2\text{O}_2$. Write the balanced equation for its combustion (reaction with O_2) to form CO_2 gas, liquid H_2O , and N_2 gas.

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- (1.) 005s (2.) 005t (3.) 005u (4.) 005v (5.) 005w (6.) 005x
(7.) 005y